T972

Quick Reference Manual

Revision: 06

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Revision History

Issue 06 (2021-02-20)

This is the 06 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
2 and 3	Deleted RealVideo related description.

Issue 05 (2020-11-05)

This is the 05 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
5.4	Added thermal resistance specification of FCCSP package.
6	Added dimension figure of FCCSP package.

Issue 04 (2020-07-20)

This is the 04 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
3	Changed eMMC specification to HS400.
5.6.2	Added description of eMMC HS400.

Issue 03 (2020-02-18)

This is the 03 release.

Compared to last version, the following contents are changed.

Section	Change Description
2 and 3	Added HDCP 2.3 description.
5.2	Changed the parameter values of VDD_EE.
5.5.2	Added a note explaining “OD 5V”.

Issue 02 (2019-10-16)

This is the 02 release.

Compared to last version, the following topics are added.

- 5.6 and its child topics.
- 7.1

Compared to last version, the following topics are changed.

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Section	Change Description
2, 3, 4.3 and 5.2	Added the description of T972 supporting LVDS.
3	Modified supported eMMC mode spec.

Issue 01 (2019-08-08)

This is the 01 release.

Compared to last version, the following topics are changed.

Section	Change Description
2, 3, 4.3 and 5.2	Deleted the description of LVDS.
5.2	Changed the spec of T_A.

Issue 0.9 (2019-05-16)

This is the preliminary 0.9 release.

Compared to last version, the following topic is added.

- 1

Compared to last version, the following topic is deleted.

- Ordering Guide

Issue 0.8 (2019-04-12)

This is the preliminary 0.8 release.

Compared to last version, the following contents are changed.

Section	Change Description
Ordering Guide	Added part number of T972–B and T972-N0E.

Issue 0.7 (2019-03-25)

This is the preliminary 0.7 release.

Compared to last version, the following contents are changed.

Section	Change Description
2 and 3	Deleted descriptions of ATSC and QAM from description and figure.

Issue 0.6 (2019-03-12)

This is the preliminary 0.6 release.

Compared to last version, the following contents are changed.

Section	Change Description
2 and 3	Changed Vulkan 1.0 to Vulkan 1.1 in GPU specification. Added 8K specifications of VP9 and H.265 in Video/Picture CODEC section.
6	Updated the dimension figure.

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Issue 0.5 (2019-01-30)

This is the preliminary 0.5 release.

Compared to last version, the following topic is added.

- Ordering Guide

Compared to last version, the following contents are changed.

Section	Change Description
4.3	Updated the pin description of the LVDS part.

Issue 0.4 (2019-01-23)

This is the preliminary 0.4 release.

Compared to last version, the following topics are changed.

Section	Change Description
5.5.3	Updated the specifications of DC Specifications - LPDDR4 mode.
5.10	Updated maximum current of VDDCPU and VDD_EE.
3	Updated specifications of Video/Picture CODEC.
5.4	Added thermal resistance specifications.

Issue 0.3 (2018-10-19)

This is the preliminary 0.3 release.

Compared to last version, the following topic is changed.

Section	Change Description
4.1	Updated the pinout diagram.

Issue 0.2 (2018-09-25)

This is the preliminary 0.2 release.

Compared to last version, the following topics are changed.

Section	Change Description
3	Modified the description of **Crypto Engine**.
6	Updated the dimension figure.

Issue 0.1 (2018-09-03)

This is the initial release.

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Contents

Revision History ii
1 **About This Document** 1
2 **General Description** 2
3 **Features Summary** 3
4 **Pinout Specification** 8
4.1 Pinout Diagram 8
4.2 Pin Order 8
4.3 Pin Description 13
4.4 Pin Multiplexing Tables 26
4.5 Signal Description 32
5 **Electrical Characteristics** 42
5.1 Absolute Maximum Ratings 42
5.2 Recommended Operating Conditions 42
5.3 Ripple Voltage Specifications 43
5.4 Thermal Resistance 43
5.5 DC Electrical Characteristics 44
5.5.1 Normal GPIO Specifications 44
5.5.2 Open Drain GPIO Specifications (For DIO_OD) 45
5.5.3 DDR3/DDR3L/DDR4/LPDDR3/LPDDR4 SDRAM Specifications 45
5.6 Timing Information 47
5.6.1 I2C Timing Specification 47
5.6.2 EMMC/SDIO Timing Specification 48
5.6.3 NAND Timing Specification 52
5.6.4 SPICC Timing Specification 55
5.6.5 SPIFC Timing Specification 55
5.6.6 Audio Timing Specification 56
5.6.7 PDM Timing Specification 60
5.6.8 UART Timing Specification 61
5.7 Recommended Oscillator Electrical Characteristics 61
5.8 Power On Config 62
5.9 Recommended Power On Sequence 63
5.10 Power Consumption 63
5.11 Storage and Baking Conditions 64
6 **Mechanical Dimensions** 66
7 **Appendix** 68
7.1 DDR Length 68

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1 About This Document

This document is applicable for T972 series SoCs, please contact your Amlogic sales representative for details.

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2 General Description

T972 is an advanced application processor designed for worldwide UHD TV applications. It integrates a powerful CPU/GPU subsystem, a best-in-class HDR image processing pipeline, a secured 8K/4K video CODEC engine with all major peripherals to form the ultimate cost-effective smart TV chip.

The main system CPU is a quad-core ARM Cortex-A55 CPU with shared L3 cache to improve system performance. In addition, the Cortex-A55 CPU includes the NEON SIMD co-processor to improve software media processing capability.

The graphic subsystem consists of two graphic engines and a flexible video/graphic output pipeline. The ARM Mali-G31 MP2 GPU handles all OpenGL ES 3.2, Vulkan 1.1 and OpenCL 2.0 graphic programs, while the 2.5D graphics processor handles additional scaling, alpha, rotation and color space conversion operations. Together, the CPU and GPU handle all operating system, network, user-interface and game related tasks.

Amlogic Video Engine (AVE-10) is a subsystem which uses dedicated hardware video decoders and encoders to offloads the Cortex-A55 CPUs from all video CODEC processing. AVE-10 is capable of decoding 4K2K resolution video within Trusted Video Path (TVP) for secured DRM applications. It supports all major video formats including MVC, MPEG-1/2/4, VC-1/WMV, AVS+, AVS2, MJPEG, H.264, H.265-10, VP9-10 and also JPEG.

The video/graphics output pipeline includes HDR10+, HDR10, HLG and Technicolor Prime HDR processing, BT.2020/ BT.2100 processing, motion compensated and motion adaptive de-interlacer, flexible programmable super scaler, local dimming and many picture enhancement filters before passing the enhanced image to the video output ports. The 8-lane V-by-one and dual-channel LVDS interface are available for UHD/FHD TV panel, and 12-lane P2P interface with internal flexible timing control module Optional for UHD TCON-less panels including CEDS, CHPI, CMPI and iSP.

3 HDMI 2.1 receiver ports plus two sets of CVBS composite analog input ports are available. The HDMI ports support HDCP 1.4/2.2/2.3 and can receive up to 4K2K HDR video.

The processor integrates the ATV demodulators which fully support worldwide analog TV standards including NTSC, PAL, and SECAM. DTV broadcasting streams can be received by the internal DTMB demodulator or the transport stream (TS) interface. The built-in three demux can process the TV streams from the serial transport stream input interface, which can connect to external tuner/demodulator. DVB Common Descrambler 1.0 is supported in addition to DES, Triple DES (TDES/3DES) and AES streaming crypto formats. An integrated ISO7816 controller is included for interfacing to external smart card.

It is optimized for low power far-field voice application. The powerful main CPU can enable top of the line audio front end and wake word algorithms. It also has built-in Voice Activity Detection (VAD) module for ultra-low power operations during system standby and full digital MIC interface including PDM, TDM and I2S up to 8 channels are available.

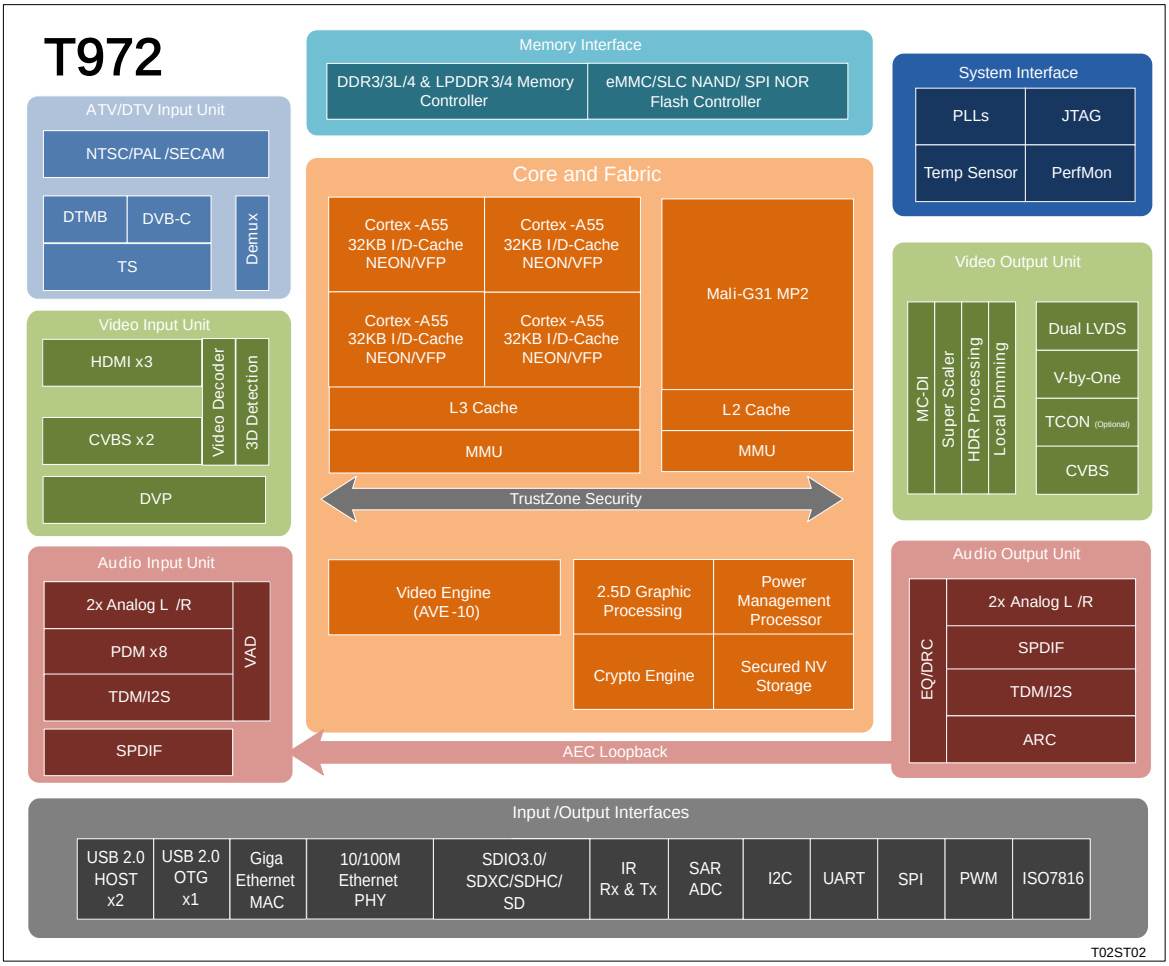
The SoC integrates rich advanced network and peripheral interfaces, including a 10/100/1000M Ethernet MAC with RGMII, 10/100M Ethernet PHY, USB 2.0 high-speed port, SDIO 3.0 controller, eMMC 5.0 controller, SLC NAND controller and multiple SDIO/SD card controllers, UART, I2C, high-speed SPI PWMs and a built-in IR blaster. The flexible and programmable QoS-based switch fabric and memory controller tie all the processing cores and peripherals together and connects to the DRAM memory bus.

Standard development environment utilizing SecureOS, Linux and GNU/GCC Android tool chain is supported. Please contact your AMLOGIC sales representative for more information.

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3 Features Summary

Figure 3-1 Features Summary



CPU Sub-system

- Quad core ARM Cortex-A55 CPU
- ARMv8.2 architecture with Neon extensions
- Unified system L3 cache
- Advanced TrustZone security system
- Application based traffic optimization using internal QoS-based switching fabrics
- CoreSight debugger support

3D Graphics Processing Unit

- ARM Mali-G31 MP2 GPU
- 4-wide warps, dual texture pipe, 2x 4-wide execution engines (EE)
- Concurrent multi-core processing
- OpenGL ES 3.2, Vulkan 1.1 and OpenCL 2.0 support

2.5D Graphics Processor

- Fast bitblt engine with dual inputs and single output

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- Programmable raster operations (ROP)
- Programmable polyphase scaling filter
- Supports multiple video formats 4:2:0, 4:2:2 and 4:4:4 and multiple pixel formats (8/16/24/32 bits graphics layer)
- Fast color space conversion
- Advanced anti-flickering filter

Crypto Engine

- AES block cipher with 128/256 bits keys, standard 16 bytes block size and streaming ECB, CBC and CTR modes
- DES/3DES block cipher with ECB and CBC modes supporting 64 bits key for DES and 192 bits key for 3DES
- Hardware key-ladder operation and DVB-CSA for transport stream encryption
- Built-in hardware True Random Number Generator (TRNG) and SHA-1/SHA-2 engine

Video/Picture CODEC

- Amlogic Video Engine (AVE-10) with dedicated hardware decoders up to 4Kx2K@75fps
- Video/Picture Decoding
  - VP9 Profile 2-10 up to 8Kx4K@24fps or 4Kx2K@60fps
  - H.265 HEVC MP-10@L5.1 up to 8Kx4K@24fps or 4Kx2K@60fps
  - AVS2-P2 Profile up to 4Kx2K@60fps
  - H.264 AVC HP@L5.1 up to 4Kx2K@30fps
  - H.264 MVC up to 1080P@60fps
  - MPEG-4 ASP@L5 up to 1080P@60fps (ISO-14496)
  - WMV/VC-1 SP/MP/AP up to 1080P@60fps
  - AVS-P16(AVS+) /AVS-P2 JiZhun Profile up to 1080P@60fps
  - MPEG-2 MP/HL up to 1080P@60fps (ISO-13818)
  - MPEG-1 MP/HL up to 1080P@60fps (ISO-11172)
  - Multiple language and multiple format sub-title video support
  - MJPEG and JPEG unlimited pixel resolution decoding (ISO/IEC-10918)
  - Supports JPEG thumbnail, scaling, rotation and transition effects
  - Supports *.mkv,*.wmv,*.mpg, *.mpeg, *.dat, *.avi, *.mov, *.iso, *.mp4 and *.jpg file formats

9th Generation Advanced Amlogic TruLife Image Engine

- Supports HDR10/10+, HLG, Technicolor Prime HDR
- Motion compensated noise reduction and 3D digital noise reduction for random noise
- Block noise, mosquito noise, spatial noise, contour noise reduction
- Motion compensated and motion adaptive de-interlacer
- Edge interpolation with low angle protection and processing
- 3:2/2:2 pulldown and Video on Film (VOF) detection and processing
- Smart sharpness with SuperScaler technology including de-contouring, de-ring, LTI, CTI, de-jaggy, peaking
- Local contrast and dynamic non-Linear contrast for detail enhancement
- 3D LUTs with 17x17x17 nodes, provide 4913 different control points, which is competent for matching calibrated displays to a target colorspace

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- High precision HSL color space based color management with low saturation protection, independent luma/hue/saturation adjustment to achieve blue/green extension, fresh tone correction, and wider gamut for video
- Video mixer: 2 video planes and 2 graphics planes
- Independent HDR re-mapping of video and graphic layer
- Local dimming control for high nits backlights

LCD Panel Output

- 8-lane V-By-One output with 1, 2, 4 regions supported, up to 4Kx2K 60Hz resolution
- Dual-channel LVDS output supporting up to 1920x1080 60Hz resolution
- Built-in (1-port 6-pair)/(2-port 3-pair) mini-LVDS output with programmable HD/FHD timing controller Optional up to 1920x1080Hz 60Hz resolution
- 12-lane CEDS/CHPI/CMPI/iSP output with programmable UHD timing controller Optional for UHD TCON-less panel, up to 4Kx2K 60Hz resolution
- Three independent Gamma table for LCD panel tuning
- Dithering logic for mapping to different LCD panel color depth

Video Input/output Interface

- 3x HDMI 2.1 receiver ports with Dynamic HDR, ARC, HDCP 1.4/2.2/2.3, 4Kx2K@60 max resolution input
- 2x CVBS 480i/576i standard definition inputs
- Supports CVBS (PAL/NTSC) bypass output
- ITU 601/656 parallel camera input supporting 8-bit RGB565, CCIR656, CCIR601, YUV422, YCbCr422

Audio Processing and Input/Output

- Supports MP3, AAC, WMA, RM, FLAC, Dolby Audio Optional, Ogg and programmable with 7.1/5.1 down-mixing
- Low-power VAD and internal AEC loopback path
- 3 built-in TDM/PCM/I2S ports with TDM/PCM mode up to 384kHz x32bits x 8ch or 96kHz x 32bits x 32ch and I2S mode up to 384kHz x 32bits x 8ch
- Digital microphone PDM voice input with programmable CIC, LPF; HPF, support up to 8 DMICs
- Built-in serial digital audio SPDIF/IEC958 output
- 2 L/R analog input channels and 2 L/R output channels
- Supports concurrent dual audio stereo channel output with combination of I2S+PCM
- Supports Audio EQ/DRC for audio speaker

TV Demodulator

- Standard compliant NTSC, NTSC-J, PAL-BG, PAL-DK1, PAL-I, PAL-DK, PAL-M, PAL-N, SECAM-DK2, SECAM-DK3, SECAM-L ATV demodulators
- Worldwide analog TV audio standard: BTSC, A2, EIA-J and NICAM
- Supports Teletext, close caption, V-chip
- DTMB/DVB-C DTV demodulators
- Build-in VIF demodulator supports low IF interface from tuner module

DTV Broadcasting Interface

- 3x Transport stream (TS) input interface with built-in demux processor for connecting to external digital TV tuner/demodulator

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- Built-in PWM, I2C and SPI interfaces to control tuner and demodulator
- Integrated ISO 7816 smart card controller

Memory and Storage Interface

- 32-bit DRAM memory interface with dual ranks and max 4GB total address space
- Compatible with JEDEC standard DDR3-2133 /DDR3L-2133 /DDR4-2666 /LPDDR3-2133 /LPDDR4-3200 SDRAM
- SDSC/SDHC/SDXC card and SDIO interface with 1-bit and 4-bit data bus width supporting spec version 2.x/3.x/4.x DS/HS modes up to UHS-I SDR104
- eMMC memory interface with 1/4/8-bit data bus width fully supporting spec version 5.0 HS400
- SLC NAND Flash controller
- Built-in 4K bits OTP memory for secured key storage

Network Interface

- IEEE 802.3 10/100/1000M Ethernet MAC with RGMII interface
- 10/100M Ethernet PHY interface
- WiFi/IEEE802.11 supporting via USB or SDIO
- Bluetooth supporting via USB or UART
- Network interface optimized for mixed WIFI and BT traffic

Integrated I/O Controllers and Interfaces

- Triple USB 2.0 high-speed USB I/O, two USB Hosts and one USB OTG
- Multiple UARTs, I2Cs and PWMs
- SPI interface
- Programmable remote control input circuitry and IR-blaster output
- Built-in 10bit SAR ADC with 4 input channels
- General Purpose IOs with built-in pull up and pull down

System, Peripherals and Misc. Interfaces

- Integrated general purpose timers, counters, DMA controllers
- 24 MHz crystal input
- Embedded debug interface using ICE/JTAG

Power Management

- Multiple internal power domains controlled by software
- Multiple sleep modes for CPU, system, DRAM, etc.
- Multiple internal PLLs to adjust the operating frequencies
- Multi-voltage I/O design for 1.8V and 3.3V

Security

- Trustzone based Trusted Execution Environment (TEE)
- Secured boot, encrypted hardware self-setup OTP, encrypted DRAM with memory integrity checker, hardware key ladder and internal control buses and storage
- Separated secure/non-secure Entropy true RNG
- Pre-region/ID memory security control and electric fence
- Hardware based Trusted Video Path (TVP), and secured contents (needs SecureOS software)
- Secured IO and secured clock

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Package

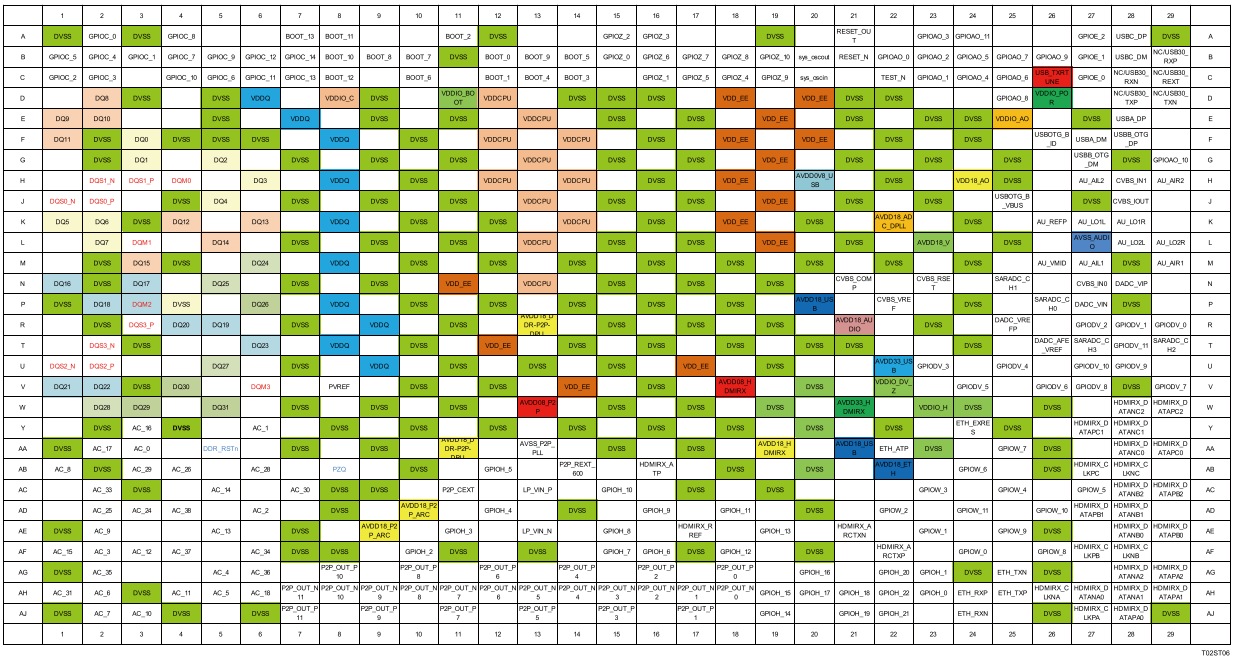
- FCBGA or FCCSP, 19 mm x 19 mm, 0.65 ball pitch, RoHS compliant

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4 Pinout Specification

4.1 Pinout Diagram

Figure 4-1 Pinout Diagram (top view)



4.2 Pin Order

BALL #	NET NAME
A1	DVSS
A2	GPIOC_0
A3	DVSS
A4	GPIOC_8
A7	BOOT_13
A8	BOOT_11
A11	BOOT_2
A12	DVSS
A15	GPIOZ_2
A16	GPIOZ_3
A19	DVSS
A21	RESET_OUT
A23	GPIOAO_3
A24	GPIOAO_11
A27	GPIOE_2

BALL #	NET NAME
A28	USBC_DP
A29	DVSS
B1	GPIOC_5
B2	GPIOC_4
B3	GPIOC_1
B4	GPIOC_7
B5	GPIOC_9
B6	GPIOC_12
B7	GPIOC_14
B8	BOOT_10
B9	BOOT_8
B10	BOOT_7
B11	DVSS
B12	BOOT_0
B13	BOOT_9

BALL #	NET NAME
B14	BOOT_5
B15	GPIOZ_0
B16	GPIOZ_6
B17	GPIOZ_7
B18	GPIOZ_8
B19	GPIOZ_10
B20	SYS_OSCOUT
B21	RESET_N
B22	GPIOAO_0
B23	GPIOAO_2
B24	GPIOAO_5
B25	GPIOAO_7
B26	GPIOAO_9
B27	GPIOE_1
B28	USBC_DM

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BALL #	NET NAME
B29	NC
C1	GPIOC_2
C2	GPIOC_3
C4	GPIOC_10
C5	GPIOC_6
C6	GPIOC_11
C7	GPIOC_13
C8	BOOT_12
C10	BOOT_6
C12	BOOT_1
C13	BOOT_4
C14	BOOT_3
C16	GPIOZ_1
C17	GPIOZ_5
C18	GPIOZ_4
C19	GPIOZ_9
C20	SYS_OSCIN
C22	TEST_N
C23	GPIOAO_1
C24	GPIOAO_4
C25	GPIOAO_6
C26	USB_TXRTUNE
C27	GPIOE_0
C28	NC
C29	NC
D2	DQ8
D3	DVSS
D5	DVSS
D6	VDDQ
D8	VDDIO_C
D9	DVSS
D11	VDDIO_BOOT
D12	VDDCPU
D14	DVSS
D15	DVSS
D16	DVSS

BALL #	NET NAME
D18	VDD_EE
D20	VDD_EE
D21	DVSS
D22	DVSS
D25	GPIOAO_8
D26	VDDIO_POR
D28	NC
D29	NC
E1	DQ9
E2	DQ10
E5	DVSS
E7	VDDQ
E9	DVSS
E11	DVSS
E13	VDDCPU
E15	DVSS
E17	DVSS
E19	VDD_EE
E21	DVSS
E23	DVSS
E24	DVSS
E25	VDDIO_AO
E27	DVSS
E28	USBA_DP
F1	DQ11
F2	DVSS
F3	DQ0
F4	DVSS
F5	DVSS
F6	DVSS
F8	VDDQ
F10	DVSS
F12	VDDCPU
F14	VDDCPU
F16	DVSS
F18	VDD_EE
F20	VDD_EE
F22	DVSS

BALL #	NET NAME
F24	DVSS
F26	USBOTG_B_ID
F27	USBA_DM
F28	USBB_OTG_DP
G2	DVSS
G3	DQ1
G5	DQ2
G7	DVSS
G9	DVSS
G11	DVSS
G13	VDDCPU
G15	DVSS
G17	DVSS
G19	VDD_EE
G21	DVSS
G23	DVSS
G25	DVSS
G27	USBB_OTG_DM
G28	DVSS
G29	GPIOAO_10
H2	DQS1_N
H3	DQS1_P
H4	DQM0
H6	DQ3
H8	VDDQ
H10	DVSS
H12	VDDCPU
H14	VDDCPU
H16	DVSS
H18	VDD_EE
H20	AVDD0V8_USB
H22	DVSS
H24	VDD18_AO
H25	DVSS
H27	AU_AIL2
H28	CVBS_IN1
H29	AU_AIR2

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BALL #	NET NAME
J1	DQS0_N
J2	DQS0_P
J4	DVSS
J5	DQ4
J7	DVSS
J9	DVSS
J11	DVSS
J13	VDDCPU
J15	DVSS
J17	DVSS
J19	VDD_EE
J21	DVSS
J23	DVSS
J25	USBOTG_B_VBUS
J27	DVSS
J28	CVBS_IOUT
K1	DQ5
K2	DQ6
K3	DVSS
K4	DQ12
K6	DQ13
K8	VDDQ
K10	DVSS
K12	DVSS
K14	VDDCPU
K16	DVSS
K18	VDD_EE
K20	DVSS
K22	AVDD18_ADC_DPLL
K24	DVSS
K26	AU_REFP
K27	AU_LO1L
K28	AU_LO1R
L2	DQ7
L3	DQM1
L5	DQ14
L7	DVSS
L9	DVSS

BALL #	NET NAME
L11	DVSS
L13	VDDCPU
L15	DVSS
L17	DVSS
L19	VDD_EE
L21	DVSS
L23	AVDD18_V
L25	DVSS
L27	AVSS_AUDIO
L28	AU_LO2L
L29	AU_LO2R
M2	DVSS
M3	DQ15
M4	DVSS
M6	DQ24
M8	VDDQ
M10	DVSS
M12	DVSS
M14	DVSS
M16	DVSS
M18	DVSS
M20	DVSS
M22	DVSS
M24	DVSS
M26	AU_VMID
M27	AU_AIL1
M28	DVSS
M29	AU_AIR1
N1	DQ16
N2	DVSS
N3	DQ17
N5	DQ25
N7	DVSS
N9	DVSS
N11	VDD_EE
N13	VDDCPU
N15	DVSS
N17	DVSS

BALL #	NET NAME
N19	DVSS
N21	CVBS_COMP
N23	CVBS_RSET
N25	SARADC_CH1
N27	CVBS_IN0
N28	DADC_VIP
P1	DVSS
P2	DQ18
P3	DQM2
P4	DVSS
P6	DQ26
P8	VDDQ
P10	DVSS
P12	DVSS
P14	DVSS
P16	DVSS
P18	DVSS
P20	AVDD18_USB
P22	CVBS_VREF
P24	DVSS
P26	SARADC_CH0
P27	DADC_VIN
P28	DVSS
R2	DVSS
R3	DQS3_P
R4	DQ20
R5	DQ19
R7	DVSS
R9	VDDQ
R11	DVSS
R13	AVDD18_DDR-P2P-DPLL
R15	DVSS
R17	DVSS
R19	DVSS
R21	AVDD18_AUDIO
R23	DVSS
R25	DADC_VREFP

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BALL #	NET NAME
R27	GPIODV_2
R28	GPIODV_1
R29	GPIODV_0
T2	DQS3_N
T3	DVSS
T6	DQ23
T8	VDDQ
T10	DVSS
T12	VDD_EE
T14	DVSS
T16	DVSS
T18	DVSS
T20	DVSS
T21	DVSS
T24	DVSS
T26	DADC_AFE_VREF
T27	SARADC_CH3
T28	GPIODV_11
T29	SARADC_CH2
U1	DQS2_N
U2	DQS2_P
U5	DQ27
U7	DVSS
U9	VDDQ
U11	DVSS
U13	DVSS
U15	DVSS
U17	VDD_EE
U19	DVSS
U22	AVDD33_USB
U23	GPIODV_3
U25	GPIODV_4
U27	GPIODV_10
U28	GPIODV_9
V1	DQ21
V2	DQ22
V3	DVSS

BALL #	NET NAME
V4	DQ30
V6	DQM3
V8	PVREF
V10	DVSS
V12	DVSS
V14	VDD_EE
V16	DVSS
V18	AVDD08_HDMIRX
V20	DVSS
V22	VDDIO_DV_Z
V24	GPIODV_5
V26	GPIODV_6
V27	GPIODV_8
V28	DVSS
V29	GPIODV_7
W2	DQ28
W3	DQ29
W5	DQ31
W7	DVSS
W9	DVSS
W11	DVSS
W13	AVDD08_P2P
W15	DVSS
W17	DVSS
W19	DVSS
W21	AVDD33_HDMIRX
W23	VDDIO_H
W24	DVSS
W26	DVSS
W28	HDMIRX_DATANC2
W29	HDMIRX_DATAPC2
Y2	DVSS
Y3	AC_16
Y4	DVSS
Y6	AC_1
Y8	DVSS
Y10	DVSS

BALL #	NET NAME
Y12	DVSS
Y14	DVSS
Y16	DVSS
Y18	DVSS
Y20	DVSS
Y22	DVSS
Y24	ETH_EXRES
Y25	DVSS
Y27	HDMIRX_DATAPC1
Y28	HDMIRX_DATANC1
AA1	DVSS
AA2	AC_17
AA3	AC_0
AA5	DDR_RSTn
AA7	DVSS
AA9	DVSS
AA11	AVDD18_DDR-P2P-DPLL
AA13	AVSS_P2P_PLL
AA15	DVSS
AA17	DVSS
AA19	AVDD18_HDMIRX
AA21	AVDD18_USB
AA22	ETH_ATP
AA23	DVSS
AA25	GPIOW_7
AA26	DVSS
AA28	HDMIRX_DATANC0
AA29	HDMIRX_DATAPC0
AB1	AC_8
AB2	DVSS
AB3	AC_29
AB4	AC_26
AB6	AC_28
AB8	PZQ
AB10	DVSS
AB12	GPIOH_5

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BALL #	NET NAME
AB14	P2P_REXT_600
AB16	HDMIRX_ATP
AB18	DVSS
AB20	DVSS
AB22	AVDD18_ETH
AB24	GPIOW_6
AB26	DVSS
AB27	HDMIRX_CLKPC
AB28	HDMIRX_CLKNC
AC2	AC_33
AC3	DVSS
AC5	AC_14
AC7	AC_30
AC8	DVSS
AC9	DVSS
AC11	P2P_CEXT
AC13	LP_VIN_P
AC15	GPIOH_10
AC17	DVSS
AC19	DVSS
AC23	GPIOW_3
AC25	GPIOW_4
AC27	GPIOW_5
AC28	HDMIRX_DATANB2
AC29	HDMIRX_DATAPB2
AD2	AC_25
AD3	AC_24
AD4	AC_38
AD6	AC_2
AD8	DVSS
AD10	AVDD18_P2P_ARC
AD12	GPIOH_4
AD14	DVSS
AD16	GPIOH_9
AD18	GPIOH_11
AD20	DVSS

BALL #	NET NAME
AD22	GPIOW_2
AD24	GPIOW_11
AD26	GPIOW_10
AD27	HDMIRX_DATAPB1
AD28	HDMIRX_DATANB1
AE1	DVSS
AE2	AC_9
AE5	AC_13
AE7	DVSS
AE9	AVDD18_P2P_ARC
AE11	GPIOH_3
AE13	LP_VIN_N
AE15	GPIOH_8
AE17	HDMIRX_RREF
AE19	GPIOH_13
AE21	HDMIRX_ARCTXN
AE23	GPIOW_1
AE25	GPIOW_9
AE26	DVSS
AE28	HDMIRX_DATANB0
AE29	HDMIRX_DATAPB0
AF1	AC_15
AF2	AC_3
AF3	AC_12
AF4	AC_37
AF6	AC_34
AF7	DVSS
AF8	DVSS
AF10	GPIOH_2
AF11	DVSS
AF13	DVSS
AF15	GPIOH_7
AF16	GPIOH_6
AF17	DVSS
AF18	GPIOH_12
AF20	DVSS

BALL #	NET NAME
AF22	HDMIRX_ARCTXP
AF24	GPIOW_0
AF26	GPIOW_8
AF27	HDMIRX_CLKPB
AF28	HDMIRX_CLKNB
AG1	DVSS
AG2	AC_35
AG5	AC_4
AG6	AC_36
AG8	P2P_OUT_P10
AG10	P2P_OUT_P8
AG12	P2P_OUT_P6
AG14	P2P_OUT_P4
AG16	P2P_OUT_P2
AG18	P2P_OUT_P0
AG20	GPIOH_16
AG22	GPIOH_20
AG23	GPIOH_1
AG24	DVSS
AG25	ETH_TXN
AG26	DVSS
AG28	HDMIRX_DATANA2
AG29	HDMIRX_DATAPA2
AH1	AC_31
AH2	AC_6
AH3	DVSS
AH4	AC_11
AH5	AC_5
AH6	AC_18
AH7	P2P_OUT_N11
AH8	P2P_OUT_N10
AH9	P2P_OUT_N9
AH10	P2P_OUT_N8
AH11	P2P_OUT_N7
AH12	P2P_OUT_N6
AH13	P2P_OUT_N5

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BALL #	NET NAME
AH14	P2P_OUT_N4
AH15	P2P_OUT_N3
AH16	P2P_OUT_N2
AH17	P2P_OUT_N1
AH18	P2P_OUT_N0
AH19	GPIOH_15
AH20	GPIOH_17
AH21	GPIOH_18
AH22	GPIOH_22
AH23	GPIOH_0
AH24	ETH_RXP
AH25	ETH_TXP
AH26	HDMIRX_CLKNA

BALL #	NET NAME
AH27	HDMIRX_DATANA0
AH28	HDMIRX_DATANA1
AH29	HDMIRX_DATAPA1
AJ1	DVSS
AJ2	AC_7
AJ3	AC_10
AJ4	DVSS
AJ6	DVSS
AJ7	P2P_OUT_P11
AJ9	P2P_OUT_P9
AJ11	P2P_OUT_P7
AJ13	P2P_OUT_P5
AJ15	P2P_OUT_P3

BALL #	NET NAME
AJ17	P2P_OUT_P1
AJ19	GPIOH_14
AJ21	GPIOH_19
AJ22	GPIOH_21
AJ24	ETH_RXN
AJ26	DVSS
AJ27	HDMIRX_CLKPA
AJ28	HDMIRX_DATAPA0
AJ29	DVSS

4.3 Pin Description

The pin assignments are described in the following table.

Table 4-1 Pin Name Assignments

Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIOC- refer to Table 4-2 for functional multiplex information.					
GPIOC_0	DIO	Up	General purpose input/output bank C signal 0	VDDIO_C	NC
GPIOC_1	DIO	Up	General purpose input/output bank C signal 1	VDDIO_C	NC
GPIOC_2	DIO	Up	General purpose input/output bank C signal 2	VDDIO_C	NC
GPIOC_3	DIO	Up	General purpose input/output bank C signal 3	VDDIO_C	NC
GPIOC_4	DIO	Up	General purpose input/output bank C signal 4	VDDIO_C	NC
GPIOC_5	DIO	Up	General purpose input/output bank C signal 5	VDDIO_C	NC
GPIOC_6	DIO	Up	General purpose input/output bank C signal 6	VDDIO_C	NC
GPIOC_7	DIO	Up	General purpose input/output bank C signal 7	VDDIO_C	NC
GPIOC_8	DIO	Up	General purpose input/output bank C signal 8	VDDIO_C	NC
GPIOC_9	DIO	Up	General purpose input/output bank C signal 9	VDDIO_C	NC
GPIOC_10	DIO	Up	General purpose input/output bank C signal 10	VDDIO_C	NC
GPIOC_11	DIO	Down	General purpose input/output bank C signal 11	VDDIO_C	NC
GPIOC_12	DIO	Up	General purpose input/output bank C signal 12	VDDIO_C	NC
GPIOC_13	DIO	Down	General purpose input/output bank C signal 13	VDDIO_C	NC
GPIOC_14	DIO	Up	General purpose input/output bank C signal 14	VDDIO_C	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
VDDIO_C	P	-	Power supply for GPIO bank C	-	1.8V or 3.3V
GPIOZ - refer to Table 4-3 for functional multiplex information.					
GPIOZ_0	DIO	Down	General purpose input/output bank Z signal 0	VDDIO_DV_Z	NC
GPIOZ_1	DIO	Up	General purpose input/output bank Z signal 1	VDDIO_DV_Z	NC
GPIOZ_2	DIO	Up	General purpose input/output bank Z signal 2	VDDIO_DV_Z	NC
GPIOZ_3	DIO	Up	General purpose input/output bank Z signal 3	VDDIO_DV_Z	NC
GPIOZ_4	DIO	Down	General purpose input/output bank Z signal 4	VDDIO_DV_Z	NC
GPIOZ_5	DIO	Down	General purpose input/output bank Z signal 5	VDDIO_DV_Z	NC
GPIOZ_6	DIO	Down	General purpose input/output bank Z signal 6	VDDIO_DV_Z	NC
GPIOZ_7	DIO	Down	General purpose input/output bank Z signal 7	VDDIO_DV_Z	NC
GPIOZ_8	DIO	Down	General purpose input/output bank Z signal 8	VDDIO_DV_Z	NC
GPIOZ_9	DIO	Up	General purpose input/output bank Z signal 9	VDDIO_DV_Z	NC
GPIOZ_10	DIO	Up	General purpose input/output bank Z signal 10	VDDIO_DV_Z	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	1.8V or 3.3V
BOOT- refer to Table 4-4 for functional multiplex information.					
BOOT_0	DIO	Up	General purpose input/output bank BOOT signal 0	VDDIO_BOOT	NC
BOOT_1	DIO	Up	General purpose input/output bank BOOT signal 1	VDDIO_BOOT	NC
BOOT_2	DIO	Up	General purpose input/output bank BOOT signal 2	VDDIO_BOOT	NC
BOOT_3	DIO	Up	General purpose input/output bank BOOT signal 3	VDDIO_BOOT	NC
BOOT_4	DIO	Up	General purpose input/output bank BOOT signal 4	VDDIO_BOOT	NC
BOOT_5	DIO	Up	General purpose input/output bank BOOT signal 5	VDDIO_BOOT	NC
BOOT_6	DIO	Up	General purpose input/output bank BOOT signal 6	VDDIO_BOOT	NC
BOOT_7	DIO	Up	General purpose input/output bank BOOT signal 7	VDDIO_BOOT	NC
BOOT_8	DIO	Up	General purpose input/output bank BOOT signal 8	VDDIO_BOOT	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
BOOT_9	DIO	Down	General purpose input/output bank BOOT signal 9	VDDIO_BOOT	NC
BOOT_10	DIO	Up	General purpose input/output bank BOOT signal 10	VDDIO_BOOT	NC
BOOT_11	DIO	Down	General purpose input/output bank BOOT signal 11	VDDIO_BOOT	NC
BOOT_12	DIO	Up	General purpose input/output bank BOOT signal 12	VDDIO_BOOT	NC
BOOT_13	DIO	Up	General purpose input/output bank BOOT signal 13	VDDIO_BOOT	NC
VDDIO_BOOT	P	-	Power supply for GPIO bank BOOT	-	1.8V or 3.3V
GPIOH- refer to Table 4-5 for functional multiplex information.					
GPIOH_0	DIO	Down	General purpose input/output bank H signal 0	VDDIO_H	NC
GPIOH_1	DIO	Down	General purpose input/output bank H signal 1	VDDIO_H	NC
GPIOH_2	DIO	Down	General purpose input/output bank H signal 2	VDDIO_H	NC
GPIOH_3	DIO	Down	General purpose input/output bank H signal 3	VDDIO_H	NC
GPIOH_4	DIO	Down	General purpose input/output bank H signal 4	VDDIO_H	NC
GPIOH_5	DIO	Down	General purpose input/output bank H signal 5	VDDIO_H	NC
GPIOH_6	DIO	Down	General purpose input/output bank H signal 6	VDDIO_H	NC
GPIOH_7	DIO	Down	General purpose input/output bank H signal 7	VDDIO_H	NC
GPIOH_8	DIO	Down	General purpose input/output bank H signal 8	VDDIO_H	NC
GPIOH_9	DIO	Down	General purpose input/output bank H signal 9	VDDIO_H	NC
GPIOH_10	DIO	Down	General purpose input/output bank H signal 10	VDDIO_H	NC
GPIOH_11	DIO	Down	General purpose input/output bank H signal 11	VDDIO_H	NC
GPIOH_12	DIO	Down	General purpose input/output bank H signal 12	VDDIO_H	NC
GPIOH_13	DIO	Down	General purpose input/output bank H signal 13	VDDIO_H	NC
GPIOH_14	DIO	Down	General purpose input/output bank H signal 14	VDDIO_H	NC
GPIOH_15	DIO	Down	General purpose input/output bank H signal 15	VDDIO_H	NC
GPIOH_16	OD5V	-	General purpose input/output bank H signal 16	VDD18_AO	NC
GPIOH_17	DIO	Z	General purpose input/output bank H signal 17	VDDIO_H	NC
GPIOH_18	DIO	Z	General purpose input/output bank H signal 18	VDDIO_H	NC
GPIOH_19	DIO	Z	General purpose input/output bank H signal 19	VDDIO_H	NC
GPIOH_20	DIO	Z	General purpose input/output bank H signal 20	VDDIO_H	NC
GPIOH_21	DIO	Z	General purpose input/output bank H signal 21	VDDIO_H	NC
GPIOH_22	DIO	Z	General purpose input/output bank H signal 22	VDDIO_H	NC
VDDIO_H	P	-	Power supply for GPIO bank H	-	1.8V or 3.3V
GPIOAO- refer to Table 4-6 for functional multiplex information.					

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIOAO_0	DIO	Up	General purpose input/output bank AO signal 0	VDDIO_AO	NC
GPIOAO_1	DIO	Up	General purpose input/output bank AO signal 1	VDDIO_AO	NC
GPIOAO_2	DIO	Up	General purpose input/output bank AO signal 2	VDDIO_AO	NC
GPIOAO_3	DIO	Down	General purpose input/output bank AO signal 3	VDDIO_AO	NC
GPIOAO_4	DIO	Down	General purpose input/output bank AO signal 4	VDDIO_AO	NC
GPIOAO_5	DIO	Up	General purpose input/output bank AO signal 5	VDDIO_AO	NC
GPIOAO_6	DIO	Down	General purpose input/output bank AO signal 6	VDDIO_AO	NC
GPIOAO_7	DIO	Down	General purpose input/output bank AO signal 7	VDDIO_AO	NC
GPIOAO_8	DIO	Down	General purpose input/output bank AO signal 8	VDDIO_AO	NC
GPIOAO_9	DIO	Down	General purpose input/output bank AO signal 9	VDDIO_AO	NC
GPIOAO_10	DIO	Z	General purpose input/output bank AO signal 10	VDDIO_AO	NC
GPIOAO_11	DIO	Down	General purpose input/output bank AO signal 11	VDDIO_AO	NC
TEST_N	DIO	Up	SOC test pin and general purpose input/output bank AO signal 12. Should be pulled up during normal power-on.	VDDIO_AO	NC
RESET_N	Input	Down	System reset input	VDDIO_AO	To RESET_N
VDDIO_AO	P	-	Power supply for GPIO bank AO	VDDIO_AO	1.8V or 3.3V
GPIOE- refer to Table 4-7 for functional multiplex information.					
GPIOE_0	DIO	Z	General purpose input/output bank E signal 0	VDD18_AO	NC
GPIOE_1	DIO	Z	General purpose input/output bank E signal 1	VDD18_AO	NC
GPIOE_2	DIO	Z	General purpose input/output bank E signal 2	VDD18_AO	NC
VDD18_AO	P	-	Power supply for GPIO bank E, SARADC, SYS_SOC, EFUSE, and IOVREF	-	VDD18_AO
GPIODV- refer to Table 4-8 for functional multiplex information.					
GPIODV_0	DIO	Up	General purpose input/output bank DV signal 0	VDDIO_DV_Z	NC
GPIODV_1	DIO	Up	General purpose input/output bank DV signal 1	VDDIO_DV_Z	NC
GPIODV_2	DIO	Up	General purpose input/output bank DV signal 2	VDDIO_DV_Z	NC
GPIODV_3	DIO	Down	General purpose input/output bank DV signal 3	VDDIO_DV_Z	NC
GPIODV_4	DIO	Down	General purpose input/output bank DV signal 4	VDDIO_DV_Z	NC
GPIODV_5	DIO	Down	General purpose input/output bank DV signal 5	VDDIO_DV_Z	NC
GPIODV_6	DIO	Up	General purpose input/output bank DV signal 6	VDDIO_DV_Z	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIODV_7	DIO	Up	General purpose input/output bank DV signal 7	VDDIO_DV_Z	NC
GPIODV_8	DIO	Up	General purpose input/output bank DV signal 8	VDDIO_DV_Z	NC
GPIODV_9	DIO	Up	General purpose input/output bank DV signal 9	VDDIO_DV_Z	NC
GPIODV_10	DIO	Up	General purpose input/output bank DV signal 10	VDDIO_DV_Z	NC
GPIODV_11	DIO	Up	General purpose input/output bank DV signal 11	VDDIO_DV_Z	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	1.8V or 3.3V
GPIOW- refer to Table 4-9 for functional multiplex information.					
GPIOW_0	OD5V	-	General purpose input/output bank W signal 0	VDD18_AO	NC
GPIOW_1	OD5V	-	General purpose input/output bank W signal 1	VDD18_AO	NC
GPIOW_2	OD5V	-	General purpose input/output bank W signal 2	VDD18_AO	NC
GPIOW_3	OD5V	-	General purpose input/output bank W signal 3	VDD18_AO	NC
GPIOW_4	OD5V	-	General purpose input/output bank W signal 4	VDD18_AO	NC
GPIOW_5	OD5V	-	General purpose input/output bank W signal 5	VDD18_AO	NC
GPIOW_6	OD5V	-	General purpose input/output bank W signal 6	VDD18_AO	NC
GPIOW_7	OD5V	-	General purpose input/output bank W signal 7	VDD18_AO	NC
GPIOW_8	OD5V	-	General purpose input/output bank W signal 8	VDD18_AO	NC
GPIOW_9	OD5V	-	General purpose input/output bank W signal 9	VDD18_AO	NC
GPIOW_10	OD5V	-	General purpose input/output bank W signal 10	VDD18_AO	NC
GPIOW_11	OD5V	-	General purpose input/output bank W signal 11	VDD18_AO	NC
SARADC					
SARADC_CH0	AI	-	ADC channel 0 input	VDD18_AO	NC
SARADC_CH1	AI	-	ADC channel 1 input	VDD18_AO	NC
SARADC_CH2	AI	-	ADC channel 2 input	VDD18_AO	NC
SARADC_CH3	AI	-	ADC channel 3 input	VDD18_AO	NC
VDD18_AO	P	-	Power supply for AO 1.8V	-	To 1.8V
CVBS OUT					
CVBS_COMP	A	-	CVBS external compensation capacitor connection	AVDD18_V	NC
CVBS_RSET	A	-	CVBS output strength setting resistor	AVDD18_V	NC
CVBS_VREF	A	-	CVBS reference voltage filter cap	AVDD18_V	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
CVBS_IOUT	AO	-	CVBS output	AVDD18_V	NC
AVDD18_V	P	-	Power Supply for analog 1.8V	-	To 1.8V
CVBS IN					
CVBS_IN0	AI	-	Analog CVBS input port 0	AVDD18_V	NC
CVBS_IN1	AI	-	Analog CVBS input port 1	AVDD18_V	NC
DADC_ AFE_VREF	A	-	CVBS input reference voltage	AVDD18_V	NC
DADC_ VREFP	A	-	CVBS ADC internal positive reference voltage	AVDD18_V	NC
AVDD18_ ADC_DPLL	A	-	Power supply for ADC PLL	-	1.8V
Demod					
DADC_VIN	AI	-	Demod ADC Low IF_N input	AVDD18_V	NC
DADC_VIP	AI	-	Demod ADC Low IF_P input	AVDD18_V	NC
LVDS/V-by-One/P2P-TCON					
P2P_OUT_ N0	AO	-	LVDS A/V-by-one/P2P data0 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P0	AO	-	LVDS A/V-by-one/P2P data0 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N1	AO	-	LVDS A/V-by-one/P2P data1 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P1	AO	-	LVDS A/V-by-one/P2P data1 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N2	AO	-	LVDS A/V-by-one/P2P data2 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P2	AO	-	LVDS A/V-by-one/P2P data2 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N3	AO	-	LVDS A CLK, V-by-one/P2P data3 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P3	AO	-	LVDS A CLK, V-by-one/P2P data3 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N4	AO	-	LVDS A data3, V-by-one/P2P data4 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P4	AO	-	LVDS A data3, V-by-one/P2P data4 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N5	AO	-	LVDS A data4, V-by-one/P2P data5 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P5	AO	-	LVDS A data4, V-by-one/P2P data5 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N6	AO	-	LVDS B data0, V-by-one/P2P data6 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P6	AO	-	LVDS B data0, V-by-one/P2P data6 positive output	AVDD18_ P2P_ARC	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
P2P_OUT_N7	AO	-	LVDS B data1, V-by-one/P2P data7 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P7	AO	-	LVDS B data1, V-by-one/P2P data7 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N8	AO	-	LVDS port B data2, P2P data8 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P8	AO	-	LVDS port B data2, P2P data8 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N9	AO	-	LVDS port B CLK, P2P data9 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P9	AO	-	LVDS port B CLK, P2P data9 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N10	AO	-	LVDS port B data3, P2P data10 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P10	AO	-	LVDS port B data3, P2P data10 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N11	AO	-	LVDS port B data4, P2P data11 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P11	AO	-	LVDS port B data4, P2P data11 positive output	AVDD18_P2P_ARC	NC
P2P_REXT_600	AO	-	LVDS and V-by-one reference resistor connection	AVDD18_P2P_ARC	NC
P2P_CEXT	AO	-	LVDS and P2P port test pin	AVDD18_P2P_ARC	NC
LP_VIN_P	AO	-	LVDS and P2P port test output	AVDD18_P2P_ARC	NC
LP_VIN_N	AO	-	LVDS and P2P port test output	AVDD18_P2P_ARC	NC
AVDD18_P2P_ARC	AP	-	Analog power supply 1.8V for V-by-one	-	1.8V
AVSS_P2P_PLL	AP	-	Analog ground for P2P PLL	-	GND
AVDD08_P2P	AP	-	Power supply for P2P analog 0.8V	-	0.8V
HDMI RX					
HDMIRX_ARCTXN	AO	-	Audio Return Channel output	AVDD18_P2P_ARC	NC
HDMIRX_ARCTXP	AO	-	Audio Return Channel output	AVDD18_P2P_ARC	NC
HDMIRX_RREF	A	-	HDMIRX Reference resistor connection	AVDD18_HDMI	NC
HDMIRX_CLKNA	AI	-	HDMIRX Port A TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPA	AI	-	HDMIRX Port A TMDS clock positive input	AVDD33_HDMI	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
HDMIRX_DATANA0	AI	-	HDMIRX Port A TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA0	AI	-	HDMIRX Port A TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANA1	AI	-	HDMIRX Port A TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA1	AI	-	HDMIRX Port A TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANA2	AI	-	HDMIRX Port A TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA2	AI	-	HDMIRX Port A TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_CLKNB	AI	-	HDMIRX Port B TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPB	AI	-	HDMIRX Port B TMDS clock positive input	AVDD33_HDMI	NC
HDMIRX_DATANB0	AI	-	HDMIRX Port B TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB0	AI	-	HDMIRX Port B TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANB1	AI	-	HDMIRX Port B TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB1	AI	-	HDMIRX Port B TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANB2	AI	-	HDMIRX Port B TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB2	AI	-	HDMIRX Port B TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_CLKNC	AI	-	HDMIRX Port C TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPC	AI	-	HDMIRX Port C TMDS clock positive input	AVDD33_HDMI	NC
HDMIRX_DATANC0	AI	-	HDMIRX Port C TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC0	AI	-	HDMIRX Port C TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANC1	AI	-	HDMIRX Port C TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC1	AI	-	HDMIRX Port C TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANC2	AI	-	HDMIRX Port C TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC2	AI	-	HDMIRX Port C TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_ATP	-	-	HDMIRX test port	AVDD33_HDMI	NC
AVDD33_HDMIRX	AP	-	Power supply for HDMI RX analog 3.3V	-	To 3.3V

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AVDD18_ HDMIRX	AP	-	Power supply for HDMI RX analog 1.8V	-	To 1.8V
AVDD08_ HDMIRX	AP	-	Power supply for HDMI RX analog 0.8V	-	To 0.8V
Audio					
AU_AIL1	AI	-	Audio line in, left channel of port 1	AVDD18_ AUDIO	NC
AU_AIR1	AI	-	Audio line in, right channel of port 1	AVDD18_ AUDIO	NC
AU_AIL2	AI	-	Audio line in, left channel of port 2	AVDD18_ AUDIO	NC
AU_AIR2	AI	-	Audio line in, right channel of port 2	AVDD18_ AUDIO	NC
AU_LO1L	AO	-	Audio line out, left channel of port 1	AVDD18_ AUDIO	NC
AU_LO1R	AO	-	Audio line out, right channel of port 1	AVDD18_ AUDIO	NC
AU_LO2L	AO	-	Audio line out, left channel of port 2	AVDD18_ AUDIO	NC
AU_LO2R	AO	-	Audio line out, right channel of port 2	AVDD18_ AUDIO	NC
AU_REFP	A	-	Audio ADC/DAC positive reference voltage	AVDD18_ AUDIO	NC
AU_VMID	A	-	MID voltage of audio ADC/DAC	AVDD18_ AUDIO	NC
AVDD18_ AUDIO	AP	-	Analog 1.8V for Audio CODEC	-	To 1.8V
AVSS_ Audio	AP	-	Analog ground for Audio CODEC	-	To VSS
DRAM- refer to Table 4-10 for functional multiplex information.					
AC_0	DO	-	DDR PHY address/command/control signal bit 0	VDDQ	NC
AC_1	DO	-	DDR PHY address/command/control signal bit 1	VDDQ	NC
AC_2	DO	-	DDR PHY address/command/control signal bit 2	VDDQ	NC
AC_3	DO	-	DDR PHY address/command/control signal bit 3	VDDQ	NC
AC_4	DO	-	DDR PHY address/command/control signal bit 4	VDDQ	NC
AC_5	DO	-	DDR PHY address/command/control signal bit 5	VDDQ	NC
AC_6	DO	-	DDR PHY address/command/control signal bit 6	VDDQ	NC
AC_7	DO	-	DDR PHY address/command/control signal bit 7	VDDQ	NC
AC_8	DO	-	DDR PHY address/command/control signal bit 8	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AC_9	DO	-	DDR PHY address/command/control signal bit 9	VDDQ	NC
AC_10	DO	-	DDR PHY address/command/control signal bit 10	VDDQ	NC
AC_11	DO	-	DDR PHY address/command/control signal bit 11	VDDQ	NC
AC_12	DO	-	DDR PHY address/command/control signal bit 12	VDDQ	NC
AC_13	DO	-	DDR PHY address/command/control signal bit 13	VDDQ	NC
AC_14	DO	-	DDR PHY address/command/control signal bit 14	VDDQ	NC
AC_15	DO	-	DDR PHY address/command/control signal bit 15	VDDQ	NC
AC_16	DO	-	DDR PHY address/command/control signal bit 16	VDDQ	NC
AC_17	DO	-	DDR PHY address/command/control signal bit 17	VDDQ	NC
AC_18	DO	-	DDR PHY address/command/control signal bit 18	VDDQ	NC
AC_24	DO	-	DDR PHY address/command/control signal bit 24	VDDQ	NC
AC_25	DO	-	DDR PHY address/command/control signal bit 25	VDDQ	NC
AC_26	DO	-	DDR PHY address/command/control signal bit 26	VDDQ	NC
AC_28	DO	-	DDR PHY address/command/control signal bit 28	VDDQ	NC
AC_29	DO	-	DDR PHY address/command/control signal bit 29	VDDQ	NC
AC_30	DO	-	DDR PHY address/command/control signal bit 30	VDDQ	NC
AC_31	DO	-	DDR PHY address/command/control signal bit 31	VDDQ	NC
AC_33	DO	-	DDR PHY address/command/control signal bit 33	VDDQ	NC
AC_34	DO	-	DDR PHY address/command/control signal bit 34	VDDQ	NC
AC_35	DO	-	DDR PHY address/command/control signal bit 35	VDDQ	NC
AC_36	DO	-	DDR PHY address/command/control signal bit 36	VDDQ	NC
AC_37	DO	-	DDR PHY address/command/control signal bit 37	VDDQ	NC
AC_38	DO	-	DDR PHY address/command/control signal bit 38	VDDQ	NC
DDR_RSTn	DO	-	DDR3/DDR4/LPDDR4 RSTn	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
DQ0	DIO	-	DRAM data bus bit 0	VDDQ	To DRAM
DQ1	DIO	-	DRAM data bus bit 1	VDDQ	To DRAM
DQ2	DIO	-	DRAM data bus bit 2	VDDQ	To DRAM
DQ3	DIO	-	DRAM data bus bit 3	VDDQ	To DRAM
DQ4	DIO	-	DRAM data bus bit 4	VDDQ	To DRAM
DQ5	DIO	-	DRAM data bus bit 5	VDDQ	To DRAM
DQ6	DIO	-	DRAM data bus bit 6	VDDQ	To DRAM
DQ7	DIO	-	DRAM data bus bit 7	VDDQ	To DRAM
DQ8	DIO	-	DRAM data bus bit 8	VDDQ	To DRAM
DQ9	DIO	-	DRAM data bus bit 9	VDDQ	To DRAM
DQ10	DIO	-	DRAM data bus bit 10	VDDQ	To DRAM
DQ11	DIO	-	DRAM data bus bit 11	VDDQ	To DRAM
DQ12	DIO	-	DRAM data bus bit 12	VDDQ	To DRAM
DQ13	DIO	-	DRAM data bus bit 13	VDDQ	To DRAM
DQ14	DIO	-	DRAM data bus bit 14	VDDQ	To DRAM
DQ15	DIO	-	DRAM data bus bit 15	VDDQ	To DRAM
DQ16	DIO	-	DRAM data bus bit 16	VDDQ	NC
DQ17	DIO	-	DRAM data bus bit 17	VDDQ	NC
DQ18	DIO	-	DRAM data bus bit 18	VDDQ	NC
DQ19	DIO	-	DRAM data bus bit 19	VDDQ	NC
DQ20	DIO	-	DRAM data bus bit 20	VDDQ	NC
DQ21	DIO	-	DRAM data bus bit 21	VDDQ	NC
DQ22	DIO	-	DRAM data bus bit 22	VDDQ	NC
DQ23	DIO	-	DRAM data bus bit 23	VDDQ	NC
DQ24	DIO	-	DRAM data bus bit 24	VDDQ	NC
DQ25	DIO	-	DRAM data bus bit 25	VDDQ	NC
DQ26	DIO	-	DRAM data bus bit 26	VDDQ	NC
DQ27	DIO	-	DRAM data bus bit 27	VDDQ	NC
DQ28	DIO	-	DRAM data bus bit 28	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
DQ29	DIO	-	DRAM data bus bit 29	VDDQ	NC
DQ30	DIO	-	DRAM data bus bit 30	VDDQ	NC
DQ31	DIO	-	DRAM data bus bit 31	VDDQ	NC
DQM0	DIO	-	DRAM data mask 0	VDDQ	To DRAM
DQM1	DIO	-	DRAM data mask 1	VDDQ	To DRAM
DQM2	DIO	-	DRAM data mask 2	VDDQ	NC
DQM3	DIO	-	DRAM data mask 3	VDDQ	NC
DQS0_P	DIO	-	DRAM data strobe 0	VDDQ	To DRAM
DQS0_N	DIO	-	DRAM data strobe 0 complementary	VDDQ	To DRAM
DQS1_P	DIO	-	DRAM data strobe 1	VDDQ	To DRAM
DQS1_N	DIO	-	DRAM data strobe 1 complementary	VDDQ	To DRAM
DQS2_P	DIO	-	DRAM data strobe 2	VDDQ	NC
DQS2_N	DIO	-	DRAM data strobe 2 complementary	VDDQ	NC
DQS3_P	DIO	-	DRAM data strobe 3	VDDQ	NC
DQS3_N	DIO	-	DRAM data strobe 3 complementary	VDDQ	NC
PZQ	A	-	DRAM reference pin for ZQ calibration, to GND by 240ohm	VDDQ	-
PVREF	A	-	DRAM reference voltage	VDDQ	NC
USB					
USBA_DP	AIO	-	USB 2.0 Port A host positive data signal	AVDD33_ USB	NC
USBA_DM	AIO	-	USB 2.0 Port A host negative data signal	AVDD33_ USB	NC
USBB_ OTG_DP	AIO	-	USB 2.0 Port B OTG positive data signal	AVDD33_ USB	NC
USBB_ OTG_DM	AIO	-	USB 2.0 Port B OTG negative data signal	AVDD33_ USB	NC
USBOTG_ B_ID	AI	-	USB 2.0 Port B ID detect signal, internal pull up to 1.8V	AVDD18_ USB	NC
USBOTG_ B_VBUS	AI	-	USB 2.0 Port B host cable power detection (1.8V input tolerance)	AVDD18_ USB	NC
USBC_DP	AIO	-	USB 2.0 Port C host positive data signal	AVDD33_ USB	NC
USBC_DM	AIO	-	USB 2.0 Port C host negative data signal	AVDD33_ USB	NC
USB_ TXRTUNE	AIO	-	USB 2.0 Port host output strength setting resistor	AVDD33_ USB	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AVDD33_ USB	AP	-	Power supply for USB 2.0(A&B&C Channel) analog 3.3V	-	To 3.3V
AVDD18_ USB	AP	-	Power supply for USB 2.0(A&B&C Channel) analog 1.8V	-	To 1.8V
AVDD0V8_ USB	AP	-	Power supply for USB analog 0.8V	-	To 0.8V
Ethernet					
ETH_ATP	-	-	Ethernet test port	AVDD18_ ETH	NC
ETH_ EXRES	A	-	Ethernet PHY external resistor connection	AVDD18_ ETH	NC
ETH_RXN	AIO	-	Ethernet PHY receive date negative input	AVDD18_ ETH	NC
ETH_RXP	AIO	-	Ethernet PHY receive data positive input	AVDD18_ ETH	NC
ETH_TXN	AIO	-	Ethernet PHY transmit data negative output	AVDD18_ ETH	NC
ETH_TXP	AIO	-	Ethernet PHY transmit data positive output	AVDD18_ ETH	NC
AVDD18_ ETH	AP	-	Analog 1.8V power supply for Ethernet	-	To 1.8V
System Clock & PLL					
SYS_ OSCIN	AI	-	24MHz crystal oscillator input	VDD18_AO	-
SYS_ OSCOUT	AO	-	24MHz crystal oscillator output	VDD18_AO	-
AVDD18_ DDR-P2P-DPLL	AP	-	Power supply for DPLL analog 1.8V	-	-
Power On Reset					
RESET_ OUT	AO	Down	Power on reset output	VDDIO_ POR	NC
VDDIO_ POR	AP	-	Power supply for POR	-	NC
Digital Power					
VDDCPU	P	-	Power supply for CPU (Cortex A55)	-	To VDDCP-U
VDDQ	P	-	DDR IO Power supply for DDR PHY	-	To VDDQ
VDD_EE	P	-	Power supply for GPU and core logic	-	To VDD_ EE
VDD18_AO	P	-	1.8V Power supply for Always On Domain	-	To VDD18_ AO
VDDIO_ BOOT	P	-	Power supply for GPIO bank BOOT	-	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
VDDIO_C	P	-	Power supply for GPIO bank C	-	1.8V or 3.3V
VDDIO_H	P	-	Power supply for GPIO bank H	-	NC
VDDIO_AO	P	-	Power supply for GPIO bank AO	-	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	NC
Digital Ground					
DVSS	P	-	Digital power ground	-	To GND

Abbreviations:

- DI = Digital input pin
- DO = Digital output pin
- DIO = Digital input/output pin
- OD5V = 5V input tolerant open drain (OD) output pin, need external pull up
- A = Analog setting or filtering pin
- AI = Analog input pin
- AO = Analog output pin
- AIO = Analog input/output pin
- P = Power pin
- AP = Analog power pin
- NC = No connection
- Up = Pull-Up
- Down = Pull-down
- Z = High-Z

4.4 Pin Multiplexing Tables

Multiple usage pins are used to conserve pin consumption for different features. The device can be used in many different applications but each application will not utilize all the onchip features. As a result, some of the features share the same pin. Most of the multiple usage pins can be used as a GPIO pin as well.

Table 4-2 GPIOC_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOC_0	SDCARD_D0		TDMB_D0	TSout_CLK	SPI_MISO_B	
GPIOC_1	SDCARD_D1		TDMB_D1	TSout_SOP	SPI_MOSI_B	
GPIOC_2	SDCARD_D2		TDMB_D2	TSout_VALID	SPI_CLK_B	
GPIOC_3	SDCARD_D3		TDMB_D3	TSout_D0	SPI_SS0_B	
GPIOC_4	SDCARD_CLK		TDMB_SCLK	TSout_D1	SPI_SS1_B	
GPIOC_5	SDCARD_CMD		TDMB_FS	TSout_D2	SPI_SS2_B	

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Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOC_6	UART_EE_A_TX	PDM_DCLK	TDMB_DIN0	TSout_D3	DTV_IF_AGC	ATV_IF_AGC
GPIOC_7	UART_EE_A_RX	PDM_DIN0	TDMB_DIN1	TSout_D4	DTV_IF_AGC	DTV_RF_AGC
GPIOC_8	UART_EE_A_CTS	PDM_DIN1	TDMB_DIN2	TSout_D5	I2C_EE_M0_SCL	
GPIOC_9	UART_EE_A_RTS	PDM_DIN2	TDMB_DIN3	TSout_D6	I2C_EE_M0_SDA	
GPIOC_10	PWM_B	PDM_DIN3	MCLK_1	TSout_D7		
GPIOC_11	SPI_MISO_A	TDMB_D1	TDMB_DIN1			
GPIOC_12	SPI_MOSI_A	TDMB_D0	TDMB_DIN0	WIFI_BEACON		
GPIOC_13	SPI_CLK_A	TDMB_SCLK	TDMB_SLV_SCLK	I2C_EE_M3_SCL		
GPIOC_14	SPI_SS0_A	TDMB_FS	TDMB_SLV_FS	I2C_EE_M3_SDA		

Table 4-3 GPIOZ_x Multi-Function Pin

Pin Name	Fun1	Func2	Func3	Func4
GPIOZ_0	MCLK_0	PDM_DIN1		
GPIOZ_1	TDMA_SCLK	TDMA_SLV_SCLK	I2C_EE_M1_SDA	
GPIOZ_2	TDMA_FS	TDMA_SLV_FS	I2C_EE_M1_SCL	
GPIOZ_3	TDMA_D0			PWM_A
GPIOZ_4	TDMA_D1	PDM_DIN2	WIFI_BEACON	PWM_B
GPIOZ_5	TDMA_D2		PWM_VS	PWM_C
GPIOZ_6	TDMA_D3		PWM_VS	PWM_D
GPIOZ_7	SPDIF_OUT	TDMA_DIN0	PDM_DCLK	
GPIOZ_8	MCLK_1	TDMA_DIN1	PDM_DIN0	
GPIOZ_9	I2C_EE_M2_SCL	TDMA_DIN2	PDM_DIN1	PWM_E
GPIOZ_10	I2C_EE_M2_SDA	TDMA_DIN3	PDM_DIN2	PWM_F

Table 4-4 BOOT_x Multi-Function Pin

Pin Name	Func1	Func2	Func3
BOOT_0	EMMC_D0		
BOOT_1	EMMC_D1		
BOOT_2	EMMC_D2		
BOOT_3	EMMC_D3		SPIF_HOLD
BOOT_4	EMMC_D4		SPIF_MO
BOOT_5	EMMC_D5		SPIF_MI

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Pin Name	Func1	Func2	Func3
BOOT_6	EMMC_D6		SPIF_CLK
BOOT_7	EMMC_D7		SPIF_WP
BOOT_8	EMMC_CLK	NAND_WEN_CLK	
BOOT_9		NAND_ALE	
BOOT_10	EMMC_CMD	NAND_REN_WR	
BOOT_11	EMMC_DS	NAND_CLE	
BOOT_12		NAND_CE0	
BOOT_13			SPIF_CS

Table 4-5 GPIOH_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOH_0	TCON_0	SPI_SS1_A	HSYNC	ETH_ACT_LED	SPDIF_OUT	I2C_EE_M3_SCL
GPIOH_1	TCON_1	SPI_SS2_A	VSYNC	ETH_LINK_LED		I2C_EE_M3_SDA
GPIOH_2	TCON_2	ETH_MDIO	BT656_A_CLK	TCON_SFC		
GPIOH_3	TCON_3	ETH_MDC	BT656_A_VS			
GPIOH_4	TCON_4	ETH_RGMII_RX_CLK	BT656_A_HS	TSin_D0_B	MCLK_0	
GPIOH_5	TCON_5	ETH_RX_DV	BT656_A_DIN0	TSin_CLK_B	TDMA_SCLK	
GPIOH_6	TCON_6	ETH_RXD0	BT656_A_DIN1	TSin_SOP_B	TDMA_FS	
GPIOH_7	TCON_7	ETH_RXD1	BT656_A_DIN2	TSin_VALID_B	TDMA_D0	
GPIOH_8	TCON_8	ETH_RXD2_RGMII	BT656_A_DIN3	TSin_D0_C	TDMA_D1	
GPIOH_9	TCON_9	ETH_RXD3_RGMII	BT656_A_DIN4	TSin_CLK_C	TDMA_D2	
GPIOH_10	TCON_10	ETH_RGMII_TX_CLK	BT656_A_DIN5	TSin_SOP_C	TDMA_D3	
GPIOH_11	TCON_11	ETH_TXEN	BT656_A_DIN6	TSin_VALID_C	TDMA_DIN0	
GPIOH_12	TCON_12	ETH_TXD0	BT656_A_DIN7		TDMA_DIN1	
GPIOH_13	TCON_13	ETH_TXD1	CLK12_24	PWM_C	TDMA_DIN2	
GPIOH_14	TCON_14	ETH_TXD2_RGMII	3D_SYNC_OUT		TDMA_DIN3	
GPIOH_15	TCON_15	ETH_TXD3_RGMII	VX1_HTPDN			
GPIOH_16	TCON_LOCK	TCON_SFC	VX1_LOCKN			
GPIOH_17	TCON_SPI_MO	SPI_MISO_A		UART_EE_B_TX	GEN_CLK_EE	

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Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOH_18	TCON_SPI_MI	SPI_MOSI_A		UART_EE_B_RX		
GPIOH_19	TCON_SPI_CLK	SPI_CLK_A	I2C_EE_M2_SCL	UART_EE_B_CTS	I2C_EE_M3_SCL	
GPIOH_20	TCON_SPI_SS	SPI_SS0_A	I2C_EE_M2_SDA	UART_EE_B_RTS	I2C_EE_M3_SDA	
GPIOH_21	I2C_EE_M1_SCL				GEN_CLK_EE	
GPIOH_22	I2C_EE_M1_SDA					

Table 4-6 GPIOAO_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5
GPIOAO_0	UART_AO_A_TX				
GPIOAO_1	UART_AO_A_RX				
GPIOAO_2	I2C_AO_M0_SCL	UART_AO_B_TX	I2C_AO_S0_SCL		
GPIOAO_3	I2C_AO_M0_SDA	UART_AO_B_RX	I2C_AO_S0_SDA		
GPIOAO_4	IR_REMOTE_OUT	CLK_32K_IN	PWMAO_C	PWMAO_C_HIZ	
GPIOAO_5	IR_REMOTE_INPUT		PWMAO_D		
GPIOAO_6	JTAG_A_CLK	SPDIF_OUT	PWMAO_C		
GPIOAO_7	JTAG_A_TMS	SPDIF_IN	PWMAO_C_HIZ		
GPIOAO_8	JTAG_A_TDI		UART_AO_B_TX		
GPIOAO_9	JTAG_A_TDO	IR_REMOTE_OUT	UART_AO_B_RX		
GPIOAO_10	AO_CEC_A	AO_CEC_B	PWMAO_D		
GPIOAO_11		PWMAO_A_HIZ	PWMAO_A	GEN_CLK_EE	GEN_CLK_AO

Table 4-7 GPIOE_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4
GPIOE_0	UART_AO_A_CTS	UART_AO_B_CTS	PWMAO_B	I2C_AO_M0_SCL
GPIOE_1	UART_AO_A_RTS	UART_AO_B_RTS	PWMAO_D	I2C_AO_M0_SDA
GPIOE_2	CLK12_24	CLK25_EE	PWM_A	

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Table 4-8 GPIODV_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIODV_0	I2C_EE_M0_SDA	TCON_SLV_SDA	TSin_D7_A			
GPIODV_1	I2C_EE_M0_SCL	TCON_SLV_SCK	TSin_D6_A			
GPIODV_2	DTV_IF_AGC	ATV_IF_AGC	TSin_D5_A	TCON_AGING	PWM_C	
GPIODV_3	DTV_IF_AGC	DTV_RF_AGC	TSin_D4_A		PWM_D	
GPIODV_4	SPDIF_OUT	TCON_FLICKER	TSin_D3_A	ISO7816_CLK		
GPIODV_5	SPDIF_IN		TSin_D2_A	ISO7816_DATA		
GPIODV_6	TDMC_D3	TDMC_DIN3	TSin_D1_A	PWM_E		
GPIODV_7	TDMC_SCLK	TDMC_SLV_SCLK	TSin_D0_A	SPI_MISO_B	UART_EE_C_TX	
GPIODV_8	TDMC_FS	TDMC_SLV_FS	TSin_CLK_A	SPI_MOSI_B	UART_EE_C_RX	
GPIODV_9	TDMC_D0	TDMC_DIN0	TSin_SOP_A	SPI_CLK_B	UART_EE_C_CTS	I2C_EE_M3_SCL
GPIODV_10	TDMC_D1	TDMC_DIN1	TSin_VALID_A	SPI_SS0_B	UART_EE_C_RTS	I2C_EE_M3_SDA
GPIODV_11	TDMC_D2	TDMC_DIN2	SPDIF_OUT	SPI_SS1_B	PWM_F	

Table 4-9 GPIOW_x Multi-Function Pin

Pin Name	Func1	Func2
GPIOW_0	HDMIRX_HPD_A	
GPIOW_1	HDMIRX_5VDET_A	
GPIOW_2	HDMIRX_SDA_A	UART_AO_A_TX
GPIOW_3	HDMIRX_SCL_A	UART_AO_A_RX
GPIOW_4	HDMIRX_HPD_C	
GPIOW_5	HDMIRX_5VDET_C	
GPIOW_6	HDMIRX_SDA_C	UART_AO_A_TX
GPIOW_7	HDMIRX_SCL_C	UART_AO_A_RX
GPIOW_8	HDMIRX_HPD_B	
GPIOW_9	HDMIRX_5VDET_B	
GPIOW_10	HDMIRX_SDA_B	UART_AO_A_TX
GPIOW_11	HDMIRX_SCL_B	UART_AO_A_RX

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Table 4-10 DDR AC Multi-Function Pin

Pin Name	LPDDR3	LPDDR4	DDR3	DDR4
AC_0	CKEA0	CKEA0	CKE0	CKE0
AC_1	CKEA1	CKEA1	CKE1	CKE1
AC_2	CSA0	CSA0	CS_N0	CS_N0
AC_3	CSA1	CSA1	NC	C0
AC_4	CLKA_T	CLKA_T	BA2	ACT_N
AC_5	CLKA_C	CLKA_C	RAS_N	WE_N
AC_6	NC	NC	A7	BA0
AC_7	NC	NC	A5	BG0
AC_8	CAA2	CAA1	A10	A10
AC_9	CAA7	CAA0	A14	A1
AC_10	CAA1	CAA4	BA0	A8
AC_11	CAA4	CAA5	CAS_N	A6
AC_12	CAA5	CAA2	A9	A2
AC_13	CAA6	CAA3	A13	A11
AC_14	CAA0	NC	WE_N	A4
AC_15	CAA3	NC	A11	A13
AC_16	CAA9	NC	CLK0_T	CLK0_T
AC_17	CAA8	NC	CLK0_C	CLK0_C
AC_18	ODTA	NC	NC	NC
AC_24	NC	CLKB_T	A6	A7
AC_25	NC	CLKB_C	A4	A5
AC_26	NC	NC	A1	A12
AC_28	NC	CAB2	A8	BA1
AC_29	NC	CAB0	BA1	RAS_N
AC_30	NC	CAB5	A15	A0
AC_31	NC	CAB4	A2	A9
AC_33	NC	CAB3	A12	CAS_N
AC_34	NC	NC	A3	BG1
AC_35	NC	NC	A0	A3
AC_36	NC	NC	ODT0	ODT0
AC_37	NC	NC	ODT1	ODT1
AC_38	NC	NC	CS_N1	CS_N1
DDR_RSTN	NC	DDR_RSTN	DDR_RSTN	DDR_RSTN
PVREF	PVREF	PVREF	PVREF	PVREF
PZQ	PZQ	PZQ	PZQ	PZQ

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4.5 Signal Description

Table 4-11 SD Card Interface Signal Description

Signal Name	Type	Description
SDCARD_D0	DIO	SD Card data bus bit 0 signal
SDCARD_D1	DIO	SD Card data bus bit 1 signal
SDCARD_D2	DIO	SD Card data bus bit 2 signal
SDCARD_D3	DIO	SD Card data bus bit 3 signal
SDCARD_CLK	DO	SD Card clock signal
SDCARD_CMD	DIO	SD Card command signal

Table 4-12 Clock Interface Signal Description

Signal Name	Type	Description
CLK_32K_IN	DI	32KHz clock input
CLK12_24	DO	12MHz/24MHZ clock output
CLK25_EE	DO	25MHz clock output

Table 4-13 UART Interface Signal Description

Signal Name	Type	Description
UART_AO_A_TX	DO	UART Port A data output in AO domain
UART_AO_A_RX	DI	UART Port A data input in AO domain
UART_AO_A_CTS	DI	UART Port A Clear To Send Signal in AO domain
UART_AO_A_RTS	DO	UART Port A Ready To Send Signal in AO domain
UART_AO_B_TX	DO	UART Port B data output in AO domain
UART_AO_B_RX	DI	UART Port B data input in AO domain
UART_AO_B_CTS	DI	UART Port B Clear To Send Signal in AO domain
UART_AO_B_RTS	DO	UART Port B Ready To Send Signal in AO domain
UART_EE_A_TX	DO	UART Port A data output in EE domain
UART_EE_A_RX	DI	UART Port A data input in EE domain
UART_EE_A_CTS	DI	UART Port A Clear To Send Signal in EE domain
UART_EE_A_RTS	DO	UART Port A Ready To Send Signal in EE domain
UART_EE_B_TX	DO	UART Port B data output in EE domain
UART_EE_B_RX	DI	UART Port B data input in EE domain
UART_EE_B_CTS	DI	UART Port B Clear To Send Signal in EE domain
UART_EE_B_RTS	DO	UART Port B Ready To Send Signal in EE domain

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Signal Name	Type	Description
UART_EE_C_TX	DO	UART Port C data output in EE domain
UART_EE_C_RX	DI	UART Port C data input in EE domain
UART_EE_C_CTS	DI	UART Port C Clear To Send Signal in EE domain
UART_EE_C_RTS	DO	UART Port C Ready To Send Signal in EE domain

Table 4-14 ISO7816 Interface Signal Description

Signal Name	Type	Description
ISO7816_DATA	DIO	ISO7816 data signal
ISO7816_CLK	DO	ISO7816 clock signal

Table 4-15 TS In Interface Signal Description

Signal Name	Type	Description
TSin_D0_A	DI	TS input port A data bus bit 0 (LSB)/ port A TS Serial data input
TSin_D1_A	DI	TS input port A data bus bit 1
TSin_D2_A	DI	TS input port A data bus bit 2
TSin_D3_A	DI	TS input port A data bus bit 3
TSin_D4_A	DI	TS input port A data bus bit 4
TSin_D5_A	DI	TS input port A data bus bit 5
TSin_D6_A	DI	TS input port A data bus bit 6
TSin_D7_A	DI	TS input port A data bus bit 7(MSB)
TSin_CLK_A	DI	TS input port A clock
TSin_SOP_A	DI	TS input port A start of stream signal
TSin_VALID_A	DI	TS input port A data valid signal
TSin_D0_B	DI	TS input port B data bus bit 0 (LSB) /port B TS Serial data input
TSin_CLK_B	DI	TS input port B clock
TSin_SOP_B	DI	TS input port B start of stream signal
TSin_VALID_B	DI	TS input port B data valid signal
TSin_D0_C	DI	TS input port C data bus bit 0 (LSB) /port C TS Serial data input
TSin_CLK_C	DI	TS input port C clock
TSin_SOP_C	DI	TS input port C start of stream signal
TSin_VALID_C	DI	TS input port C data valid signal

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Table 4-16 TS Out Interface Signal Description

Signal Name	Type	Description
TSout_D0	DO	TS output data bus bit 0 (LSB)
TSout_D1	DO	TS output data bus bit 1
TSout_D2	DO	TS output data bus bit 2
TSout_D3	DO	TS output data bus bit 3
TSout_D4	DO	TS output data bus bit 4
TSout_D5	DO	TS output data bus bit 5
TSout_D6	DO	TS output data bus bit 6
TSout_D7	DO	TS output data bus bit 7 (MSB)
TSout_CLK	DO	TS output clock
TSout_SOP	DO	TS output start of stream signal
TSout_VALID	DO	TS output date valid signal

Table 4-17 PWM Interface Signal Description

Signal Name	Type	Description
PWM_A	DO	PWM channel A output signal
PWM_B	DO	PWM channel B output signal
PWM_C	DO	PWM channel C output signal
PWM_D	DO	PWM channel D output signal
PWM_E	DO	PWM channel E output signal
PWM_F	DO	PWM channel F output signal
PWM_VS	DO	PWM VSYNC output signal
PWMAO_A	DO	PWM A output signal in Always On domain
PWMAO_B	DO	PWM B output signal in Always On domain
PWMAO_C	DO	PWM C output signal in Always On domain
PWMAO_D	DO	PWM D output signal in Always On domain
PWMAO_A_HIZ	DO	Extended HiZ function of PWMAO_A
PWMAO_C_HIZ	DO	Extended HiZ function of PWMAO_C

Table 4-18 I2C Interface Signal Description

Signal Name	Type	Description
I2C_AO_M0_SCL	DO	I2C bus port 0 clock output, Master mode, in AO domain
I2C_AO_M0_SDA	DIO	I2C bus port 0 data input/output, Master mode, in AO domain
I2C_AO_S0_SCL	DI	I2C bus port 0 clock input, Slave mode, in AO domain
I2C_AO_S0_SDA	DIO	I2C bus port 0 data input/output, Slave mode, in AO domain

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Signal Name	Type	Description
I2C_EE_M0_SCL	DO	I2C bus port 0 clock output, Master mode, in EE domain
I2C_EE_M0_SDA	DIO	I2C bus port 0 data input/output, Master mode, in EE domain
I2C_EE_M1_SCL	DO	I2C bus port 1 clock output, Master mode, in EE domain
I2C_EE_M1_SDA	DIO	I2C bus port 1 data input/output, Master mode, in EE domain
I2C_EE_M2_SCL	DO	I2C bus port 2 clock output, Master mode, in EE domain
I2C_EE_M2_SDA	DIO	I2C bus port 2 data input/output, Master mode, in EE domain
I2C_EE_M3_SCL	DO	I2C bus port 3 clock output, Master mode, in EE domain
I2C_EE_M3_SDA	DIO	I2C bus port 3 data input/output, Master mode, in EE domain

Table 4-19 TCON Interface Signal Description

Signal Name	Type	Description
TCON_0	DO	TCON control signal output
TCON_1	DO	TCON control signal output
TCON_2	DO	TCON control signal output
TCON_3	DO	TCON control signal output
TCON_4	DO	TCON control signal output
TCON_5	DO	TCON control signal output
TCON_6	DO	TCON control signal output
TCON_7	DO	TCON control signal output
TCON_8	DO	TCON control signal output
TCON_9	DO	TCON control signal output
TCON_10	DO	TCON control signal output
TCON_11	DO	TCON control signal output
TCON_12	DO	TCON control signal output
TCON_13	DO	TCON control signal output
TCON_14	DO	TCON control signal output
TCON_15	DO	TCON control signal output
TCON_LOCK	DIO	Feedback signal from source driver
TCON_SPI_MO	DIO	TCON SPI master output
TCON_SPI_MI	DI	TCON SPI master input
TCON_SPI_CLK	DO	TCON SPI Serial Clock
TCON_SPI_SS	DO	TCON SPI Slave Select
TCON_FLICKER	DI	Test flicker input signal
TCON_AGING	DI	TCON AGING mode input
TCON_SFC	DO	Dedicated USIT signal

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Table 4-20 eMMC Interface Signal Description

Signal Name	Type	Description
EMMC_D0	DIO	eMMC/NAND data bus bit 0 signal
EMMC_D1	DIO	eMMC/NAND data bus bit 1 signal
EMMC_D2	DIO	eMMC/NAND data bus bit 2 signal
EMMC_D3	DIO	eMMC/NAND data bus bit 3 signal
EMMC_D4	DIO	eMMC/NAND data bus bit 4 signal
EMMC_D5	DIO	eMMC/NAND data bus bit 5 signal
EMMC_D6	DIO	eMMC/NAND data bus bit 6 signal
EMMC_D7	DIO	eMMC/NAND data bus bit 7 signal
EMMC_CLK	DO	eMMC clock signal
EMMC_CMD	DIO	eMMC command signal
EMMC_DS	DI	eMMC data strobe

Table 4-21 NAND Signal Description

Signal Name	Type	Description
NAND_ALE	DO	NAND address latch enable
NAND_CE0	DO	NAND chip enable 0
NAND_CLE	DO	NAND command latch enable
NAND_REN_WR	DO	NAND read enable or write/read
NAND_WEN_CLK	DO	NAND write enable or clock

Table 4-22 HDMI Interface Signal Description

Signal Name	Type	Description
HDMIRX_HPD_A	DO	HDMI RX port A hot plug in signal output
HDMIRX_5VDET_A	DI	HDMI RX port A 5V power detection
HDMIRX_SDA_A	DIO	HDMI RX port A DDC_I2C interface data signal
HDMIRX_SCL_A	DI	HDMI RX port A DDC_I2C interface clock signal
HDMIRX_HPD_B	DO	HDMI RX port B hot plug in signal output
HDMIRX_5VDET_B	DI	HDMI RX port B 5V power detection
HDMIRX_SDA_B	DIO	HDMI RX port B DDC_I2C interface data signal
HDMIRX_SCL_B	DI	HDMI RX port B DDC_I2C interface clock signal
HDMIRX_HPD_C	DO	HDMI RX port C hot plug in signal output
HDMIRX_5VDET_C	DI	HDMI RX port C 5V power detection
HDMIRX_SDA_C	DIO	HDMI RX port C DDC_I2C interface data signal

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Signal Name	Type	Description
HDMIRX_SCL_C	DI	HDMI RX port C DDC_I2C interface clock signal
AO_CEC_A	DIO	Customer Electronics Control signal in AO domain
AO_CEC_B	DIO	2nd pin of Customer Electronics Control signal in AO domain

Table 4-23 SPIF Interface Signal Description

Signal Name	Type	Description
SPIF_CS	DO	SPIF chip select
SPIF_CLK	DO	SPIF Serial Clock
SPIF_MO	DIO	SPIF 1bit mode Output, 2/4 bit mode data I/O 0
SPIF_MI	DIO	SPIF 1bit mode Input, 2/4 bit mode data I/O 1
SPIF_WP	DIO	SPIF Write protection output, 4 bit mode data I/O 2
SPIF_HOLD	DIO	SPIF bus hold output, 4 bit mode data I/O 3

Table 4-24 SPDIF Interface Signal Description

Signal Name	Type	Description
SPDIF_IN	DI	SPDIF input signal
SPDIF_OUT	DO	SPDIF output signal

Table 4-25 SPI Interface Signal Description

Signal Name	Type	Description
SPI_MOSI_A	DIO	SPI port A master output, slave input
SPI_MISO_A	DIO	SPI port A master input, slave output
SPI_CLK_A	DIO	SPI clock, port A
SPI_SS0_A	DIO	SPI slave select 0, port A
SPI_SS1_A	DIO	SPI slave select 1, port A
SPI_SS2_A	DIO	SPI slave select 2, port A
SPI_MOSI_B	DIO	SPI port B master output, slave input
SPI_MISO_B	DIO	SPI port B master input, slave output
SPI_CLK_B	DIO	SPI clock, port B
SPI_SS0_B	DIO	SPI slave select 0, port B
SPI_SS1_B	DIO	SPI slave select 1, port B
SPI_SS2_B	DIO	SPI slave select 2, port B

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Table 4-26 Remote Interface Signal Description

Signal Name	Type	Description
IR_REMOTE_INPUT	DI	IR remote control input
IR_REMOTE_OUT	DO	IR remote control output

Table 4-27 Time Division Multiplexing Signal Description

Signal Name	Type	Description
MCLK_0	DO	Master clock output 0, for I2S master mode
MCLK_1	DO	Master clock output 1, for I2S master mode
TDMA_DIN0	DI	Data input 0 of TDM port A
TDMA_DIN1	DI	Data input 1 of TDM port A
TDMA_DIN2	DI	Data input 2 of TDM port A
TDMA_DIN3	DI	Data input 3 of TDM port A
TDMA_D0	DIO	Data input/output 0 of TDM port A
TDMA_D1	DIO	Data input/output 1 of TDM port A
TDMA_D2	DIO	Data input/output 2 of TDM port A
TDMA_D3	DIO	Data input/output 3 of TDM port A
TDMA_SCLK	DO	Bit clock output of TDM port A
TDMA_FS	DO	Frame sync output of TDM port A (Word clock of I2S)
TDMA_SLV_SCLK	DI	Bit clock input of TDM port A
TDMA_SLV_FS	DI	Frame sync input of TDM port A (Word clock of I2S)
TDMB_DIN0	DI	Data input 0 of TDM port B
TDMB_DIN1	DI	Data input 1 of TDM port B
TDMB_DIN2	DI	Data input 2 of TDM port B
TDMB_DIN3	DI	Data input 3 of TDM port B
TDMB_D0	DIO	Data input/output 0 of TDM port B
TDMB_D1	DIO	Data input/output 1 of TDM port B
TDMB_D2	DIO	Data input/output 2 of TDM port B
TDMB_D3	DIO	Data input/output 3 of TDM port B
TDMB_SCLK	DO	Bit clock output of TDM port B
TDMB_FS	DO	Frame sync output of TDM port B (Word clock of I2S)
TDMB_SLV_SCLK	DI	Bit clock input of TDM port B
TDMB_SLV_FS	DI	Frame sync input of TDM port B (Word clock of I2S)
TDMC_DIN0	DI	Data input 0 of TDM port C
TDMC_DIN1	DI	Data input 1 of TDM port C
TDMC_DIN2	DI	Data input 2 of TDM port C

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Signal Name	Type	Description
TDMC_DIN3	DI	Data input 3 of TDM port C
TDMC_D0	DIO	Data input/output 0 of TDM port C
TDMC_D1	DIO	Data input/output 1 of TDM port C
TDMC_D2	DIO	Data input/output 2 of TDM port C
TDMC_D3	DIO	Data input/output 3 of TDM port C
TDMC_SCLK	DO	Bit clock output of TDM port C
TDMC_FS	DO	Frame sync output of TDM port C (Word clock of I2S)
TDMC_SLV_SCLK	DI	Bit clock input of TDM port C
TDMC_SLV_FS	DI	Frame sync input of TDM port C (Word clock of I2S)

Table 4-28 PDM Signal Description

Signal Name	Type	Description
PDM_DIN0	DI	PDM input data 0 signal
PDM_DIN1	DI	PDM input data 1 signal
PDM_DIN2	DI	PDM input data 2 signal
PDM_DIN3	DI	PDM input data 3 signal
PDM_DCLK	DO	PDM output clock signal

Table 4-29 Tuner/Demod Interface Signal Description

Signal Name	Type	Description
DTV_IF_AGC	DO	DTV IF AGC output for external demodulator
DTV_RF_AGC	DO	DTV RF AGC output for external tuner
ATV_IF_AGC	DO	ATV IF AGC output for external demodulator

Table 4-30 3D Interface Signal Description

Signal Name	Type	Description
3D_SYNC_OUT	DO	3D sync signal output

Table 4-31 JTAG Interface Signal Description

Signal Name	Type	Description
JTAG_A_TDO	DO	JTAG data output channel A
JTAG_A_TDI	DI	JTAG data input channel A
JTAG_A_TMS	DI	JTAG Test mode select input channel A
JTAG_A_CLK	DI	JTAG Test clock input channel A

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Table 4-32 BT656 Interface Signal Description

Signal Name	Type	Description
BT656_A_DIN0	DI	BT656 input data bus bit 0
BT656_A_DIN1	DI	BT656 input data bus bit 1
BT656_A_DIN2	DI	BT656 input data bus bit 2
BT656_A_DIN3	DI	BT656 input data bus bit 3
BT656_A_DIN4	DI	BT656 input data bus bit 4
BT656_A_DIN5	DI	BT656 input data bus bit 5
BT656_A_DIN6	DI	BT656 input data bus bit 6
BT656_A_DIN7	DI	BT656 input data bus bit 7
BT656_A_CLK	DI	BT656 input Clock
BT656_A_HS	DI	BT656 input HSYNC Signal
BT656_A_VS	DI	BT656 input VSYNC Signal

Table 4-33 Ethernet Interface Signal Description

Signal Name	Type	Description
ETH_LINK_LED	DO	Ethernet link LED indicator
ETH_ACT_LED	DO	Ethernet active LED indicator
ETH_RGMII_RX_CLK	DI	Ethernet RGMII interface receive clock input
ETH_RGMII_TX_CLK	DO	Ethernet RGMII transmit clock
ETH_TXEN	DO	Ethernet RMII/RGMII Interface transmit enable
ETH_TXD3_RGMII	DO	Ethernet RGMII interface transmit data 3
ETH_TXD2_RGMII	DO	Ethernet RGMII interface transmit data 2
ETH_TXD1	DO	Ethernet RMII/RGMII interface transmit data 1
ETH_TXD0	DO	Ethernet RMII/RGMII interface transmit data 0
ETH_RX_DV	DI	Ethernet RMII/RGMII interface receive data valid signal
ETH_RXD3_RGMII	DI	Ethernet RGMII interface receive data 3
ETH_RXD2_RGMII	DI	Ethernet RGMII interface receive data 2
ETH_RXD1	DI	Ethernet RMII/RGMII interface receive data 1
ETH_RXD0	DI	Ethernet RMII/RGMII interface receive data 0
ETH_MDIO	DIO	Ethernet SMI interface management data input/output
ETH_MDC	DO	Ethernet SMI interface management clock

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Table 4-34 Other Signal Description

Signal Name	Type	Description
WIFI_BEACON	DI	General clock sync input, to sync clock of multi devices
GEN_CLK_EE	DO	General clock output for EE domain clock, for debug
GEN_CLK_AO	DO	General clock output for AO domain clock, for debug
HSYNC	DO	Horizontal synchronizing signal
VSYNC	DO	Vertical synchronizing signal
VX1_HTPDN	DI	Connection status of V-by-one receiver
VX1_LOCKN	DI	V-by-one receiver PLL lock state

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5 Electrical Characteristics

5.1 Absolute Maximum Ratings

The table below gives the absolute maximum ratings. Exposure to stresses beyond those listed in this table may result in permanent device damage, unreliability or both.

Table 5-1 Absolute Maximum Ratings

Characteristic	Value	Unit
VDDCPU Supply Voltage	1.15	V
VDD_EE Supply Voltage	1.02	V
VDDQ Supply Voltage	1.7	V
1.8V Supply Voltage	1.98	V
3.3V Supply Voltage	3.63	V
Input voltage, V_I	-0.3 ~VDDIO+0.3	V
Junction Temperature	125	°C

5.2 Recommended Operating Conditions

Table 5-2 Recommended Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDDCPU	Voltage for Cortex A55 CPU	0.70[1]	-	1.03[2]	V
VDD_EE and other 0.8V domain	Voltage for GPU & core logic	0.77	0.84	0.9	V
VDDQ	DDR3/DDR3L/DDR4/LPDDR3/LPDDR4 IO Supply Voltage	1.06	-	1.575	V
AVDD18	1.8V AVDD for HDMI, USB, AUDIO, VX1/LVDS/P2P, ETHERNET phy, DDRPLL, P2P_PLL and DPLL	1.74	1.80	1.89	V
VDD18_AO	1.8V VDD for XTAL, CVBS, SARADC, efuse and IOVREF	1.74	1.80	1.89	V
AVDD33	3.3V AVDD for USB and HDMIRX	3.15	3.3	3.45	V
VDDIO	LV mode	1.71	1.80	1.89	V
	HV mode	3.0[3]	3.3	3.45	V
T_J	Operating Junction Temperature	0	-	105[4]	°C
T_A	Operating Ambient Temperature	0	-	70	°C

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Note

1. Minimal VDDCPU/VDD_EE voltage is for sleep mode while system runs at very low speed. Higher clock will need higher voltage. Considering the power supply may have 2% deviation, the minimal voltage in actual application should not be designed to lower than the min spec value plus 0.02V.
2. Likewise, maximum VDDCPU voltage in actual application should not be higher than (max spec -0.02V). Voltage of VDDCPU will affect CPU speed. Use lower voltage when CPU runs on lower speed to save power. Recommend to use +/-1.5% or higher precision DCDC.
3. GPIO cannot work if VDDIO voltage is out of the spec of LV / HV mode. GPIO output at HV mode will be weaker & max operating speed will be lower if VDDIO are design to 3.0V. Do not design VDDIO to lower than 3.0V in HV mode.
4. For operating temperature, good heat sink may be needed to guarantee T_J < max spec.

5.3 Ripple Voltage Specifications

Table 5-3 Ripple Voltage Specifications

Power	Max Ripple	Unit	Test State
VDDCPU	40	+/-mV	Run APK StabilityTest
VDD_EE and other 0.8V domain	40	+/-mV	Run APK Basemark ES 2.0 Taiji
DDR3 VDDQ	60	+/-mV	Kernel boot
DDR3L VDDQ	60	+/-mV	Kernel boot
LPDDR3 VDDQ	40	+/-mV	Kernel boot
DDR4 VDDQ	40	+/-mV	Kernel boot
LPDDR4 VDDQ	40	+/-mV	Kernel boot
AVDD18	30	+/-mV	Kernel boot
VDD18_AO	30	+/-mV	Kernel boot
AVDD33	50	+/-mV	WIFI SCAN
VDDIO LV	60	+/-mV	Kernel boot
VDDIO HV	60	+/-mV	WIFI SCAN

Note

The ripple specifications are for reference only. Customers should perform pressure / performance / reliability tests (high / low temperature tests, wet / heat tests, functional tests, etc.) on the product to confirm system stability.

5.4 Thermal Resistance

Simulation condition: JEDEC 2s2p board 101.5 mm x 114.5 mm, natural convection, ambient temperature 25°C.

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Symbol	Parameter	Value.	Unit
Θ_{JA}	Package junction-to-ambient thermal resistance in nature convection	FCBGA: 15.30 FCCSP: 15.1	°C/Watt
Θ_{JB}	Package junction-to-board thermal resistance in nature convection	FCBGA:7.14 FCCSP: 7.2	°C/Watt
Θ_{JC}	Package junction-to-case thermal resistance in nature convection	FCBGA: 0.18 FCCSP: 4.4	°C/Watt

Note

Due to the thinness of the SoC, DRAM or capacitors placed close to SoC may prevent heatsink touching SoC top side. A special convex shape heatsink is recommended.

5.5 DC Electrical Characteristics

5.5.1 Normal GPIO Specifications

Table 5-4 Normal GPIO Specifications (For DIO)

Symbol	Parameter	Min.	Typ.	Max.	Unit
$V_{iH(VDDIO=3.3V)}$	High-level input voltage	IOVREF +0.37	-	VDDIO+0.3	V
$V_{iL(VDDIO=3.3V)}$	Low-level input voltage	-0.3	-	IOVREF-0.23	V
$V_{iH(VDDIO=1.8V)}$	High-level input voltage	IOVREF/2 +0.3	-	VDDIO+0.3	V
$V_{iL(VDDIO=1.8V)}$	Low-level input voltage	-0.3	-	IOVREF/2-0.3	V
$R_{PU/PD}$	Built-in pull up/down resistor	-	60K		ohm
IoL/IoH(DS=0)[4]	GPIO driving capability	0.5	-	-	mA
IoL/IoH(DS=1)	GPIO driving capability	2.5	-	-	mA
IoL/IoH(DS=2)	GPIO driving capability	3	-	-	mA
IoL/IoH(DS=3)	GPIO driving capability	4[1]	-	-	mA
VOH	Output high level with IoL/ IoH loading	VDDIO-0.5	-	-	V
VOL	Output low level with IoL/ IoH loading	-	-	0.4	V

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Note

1. With Minimal IoL/IoH driving capability loading, IO is garanteed to meet Vol < 0.4V or VOH > (VDDIO-0.5V) spec.
2. Maximal GPIO loading is 6mA for application such as driving LED, which does not care about Vol/Voh spec. Please set DS=3 for such application.
3. VDD18_AO supplies power to IOVREF.
4. Do not use this setting, it's too weak for most applications.

5.5.2 Open Drain GPIO Specifications (For DIO_OD)

Table 5-5 Open Drain GPIO Specifications (For DIO_OD)

Symbol	Parameter	Min.	Typ.	Max.	Unit
$V_{iH(}$OD5V)	High-level input voltage	1.5	-	5.5	V
$V_{iL(}$OD5V)	Low-level input voltage	-0.3	-	0.8	V
$R_{PU/PD}$	No built-in pull up/down resistor on OD IO	-	-	-	ohm
IoL	OD IO driving low capability	4[1]	-	6[2]	mA
VOL	Output low level with min Io loading	-	-	0.4	V

Note

1. With Minimal IoL driving capability loading, IO is garanteed to meet Vol < 0.4V spec.
2. Maximal GPIO loading is 6mA for application such as driving LED, which does not care about Vol spec.
3. The ViL/ViH of OD PAD is irrelevant to VDDIO voltage.
4. "OD 5V" means that in applications such as I2C, use a resistor greater than 1Kohm to pull it up to 5V. Do not connect the pad directly to the 5V power supply.

5.5.3 DDR3/DDR3L/DDR4/LPDDR3/LPDDR4 SDRAM Specifications

Table 5-6 Recommended Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDDQ	IO supply voltage (DDR3)	1.425	1.50	1.575	V
VDDQ	IO supply voltage (DDR3L)	1.283	1.35	1.45	V
VDDQ	IO supply voltage (DDR4)	1.14	1.20	1.26	V
VDDQ	IO supply voltage (LPDDR3)	1.14	1.2	1.30	V
VDDQ	IO supply voltage (LPDDR4)	1.06	1.1	1.17	V
Vref	Input reference supply voltage	0.49 x VDDQ	0.5 x VDDQ	0.51 x VDDQ	V

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Note

The minimal VDDQ voltage in sleep mode is defined by memory.

Table 5-7 DC specifications - DDR3/DDR3L mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VIH	DC input voltage high	Vref + 0.100	-	-	V
VIL	DC input voltage low	-	-	Vref-0.100	V
VOH	DC output logic high	0.8*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.2 x VDDQ	V
RTT	Input termination resistance to VDDQ/2	100 54 36	120 60 40	140 66 44	ohm

Table 5-8 DC specifications - DDR4 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VdlVW_total	Rx Mask voltage-p-p total	-	-	136	mv
VOH	DC output logic high	0.9 x VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1 x VDDQ	V
RTT	Input termination resistance to VDDQ	200	240	280	ohm
		100	120	140	
		67	80	93	
		50	60	70	
		42	48	56	
		34	40	46	
		28	34	40	

Table 5-9 DC Specifications - LPDDR3 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VIH	DC input voltage high	Vref + 0.100	-	-	V
VIL	DC input voltage low	-	-	Vref-0.100	V
VOH	DC output logic high	0.9*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1*VDDQ	V
RTT	Input termination resistance to VDDQ	100 200	120 240	140 280	ohm

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Table 5-10 DC Specifications - LPDDR4 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VOH	DC output logic high	0.9*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1*VDDQ	V
RTT	Input termination resistance to VDDQ	216	240	264	ohm
		108	120	132	
		72	80	88	
		54	60	66	
		43.2	48	52.8	
		36	40	44	

5.6 Timing Information

5.6.1 I2C Timing Specification

The I2C master interface Fast/Standard mode timing specifications are shown below.

Figure 5-1 I2C Interface Timing, FS mode

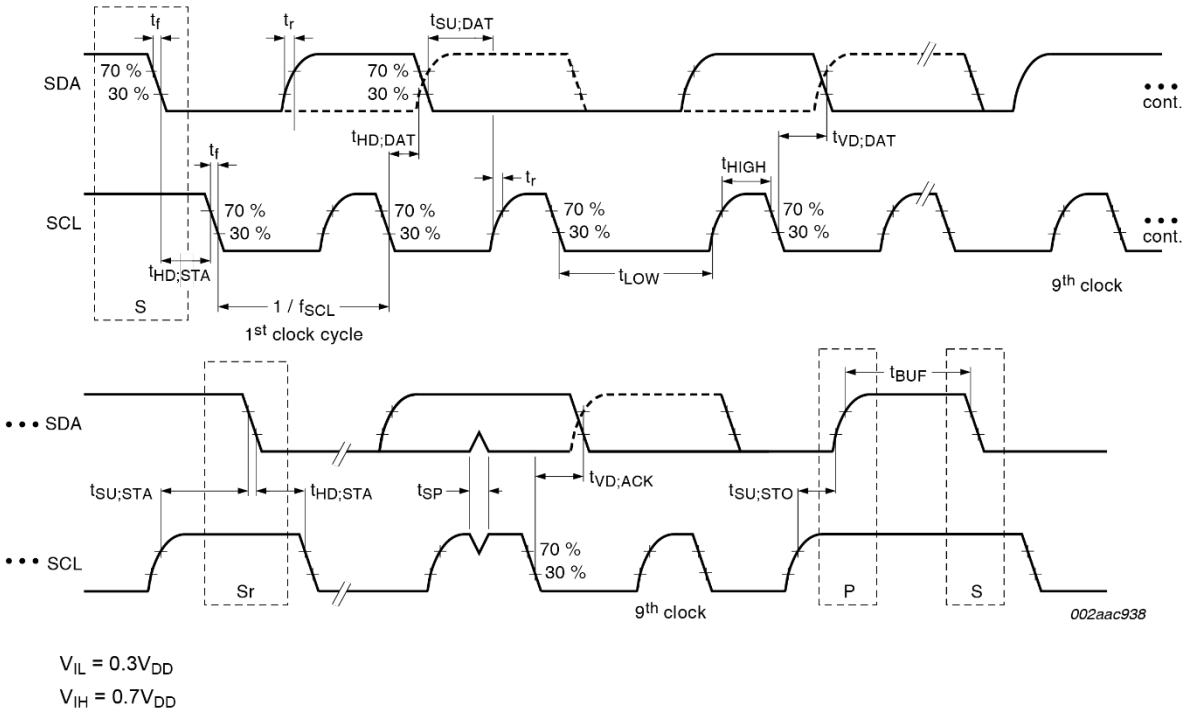


Table 5-11 I2C Interface Timing Specification, SF mode

Symbol	Parameter	Standard-mode		Fast-mode		Unit
		Min.	Max	Min	Max	
tR	Rise time of SDA and SCL signals	-	1000	-	300	ns
tF	Fall time of SDA and SCL signals	-	300	-	300	ns

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Symbol	Parameter	Standard-mode		Fast-mode		Unit
		Min.	Max	Min	Max	
fSCL	SCL clock frequency	-	100	-	400	KHz
tLOW	LOW period of the SCL clock	4.7	-	1.3	-	µs
tHIGH	HIGH period of the SCL clock	4	-	0.6	-	µs
tSu;STA	Setup time for START	4.7	-	0.6	-	µs
tSu;DAT	Setup time for SDA	250	-	100	-	ns
tSu;STO	Setup time for STOP	4	-	0.6	-	µs
tHd;STA	Hold time for START	4	-	0.6	-	µs
tHd;DAT	Hold time for SDA	0	3.45	0	0.9	µs
tBuf	Bus free time between stop and start	4.7	-	1.3	-	µs

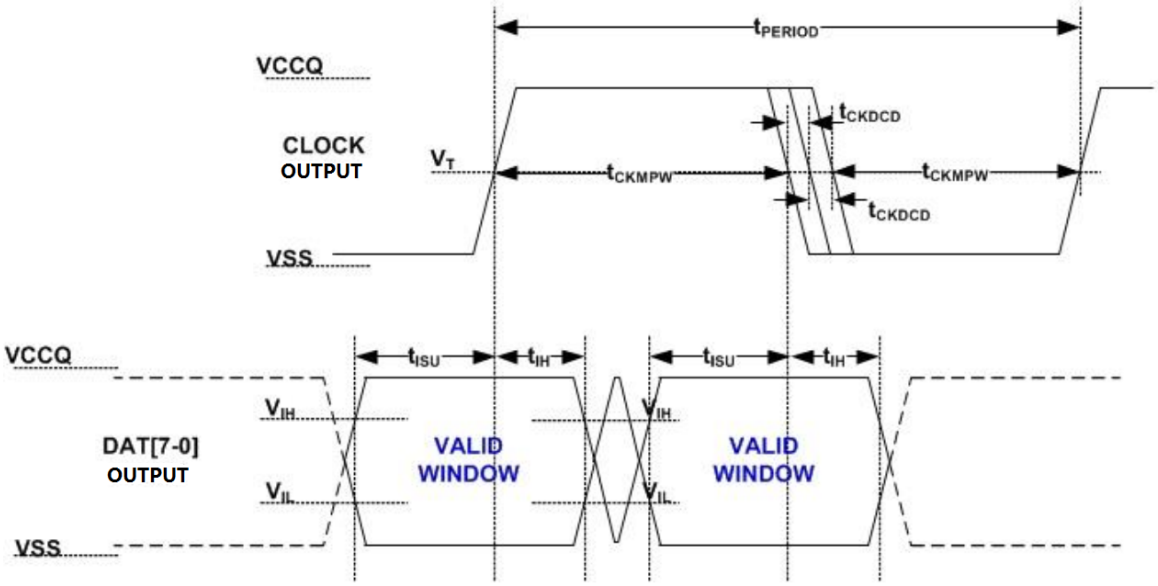
Note

Driving strength of OD pin is not adjustable.

5.6.2 EMMC/SDIO Timing Specification

Timing specification for EMMC and SD are shown as below.

Figure 5-2 EMMC HS400 Data Output Timing



NOTE V_T = 50% of V_{CCQ}, indicates clock reference point for timing measurements.

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Table 5-12 HS400 Output Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
SR	Slew rate	1.125	-	V/ns
tCKDCD	Duty cycle distortion	0	0.3	ns
tCKMPW	Minimum pulse width	2.2	-	ns
tISU	input set-up time	1.4	-	ns
tIH	input hold time	0.8	-	ns
tISUddr	input set-up time	0.4	-	ns
tIHddr	input hold time	0.4	-	ns

Figure 5-3 EMMC HS200 Data Output Timing

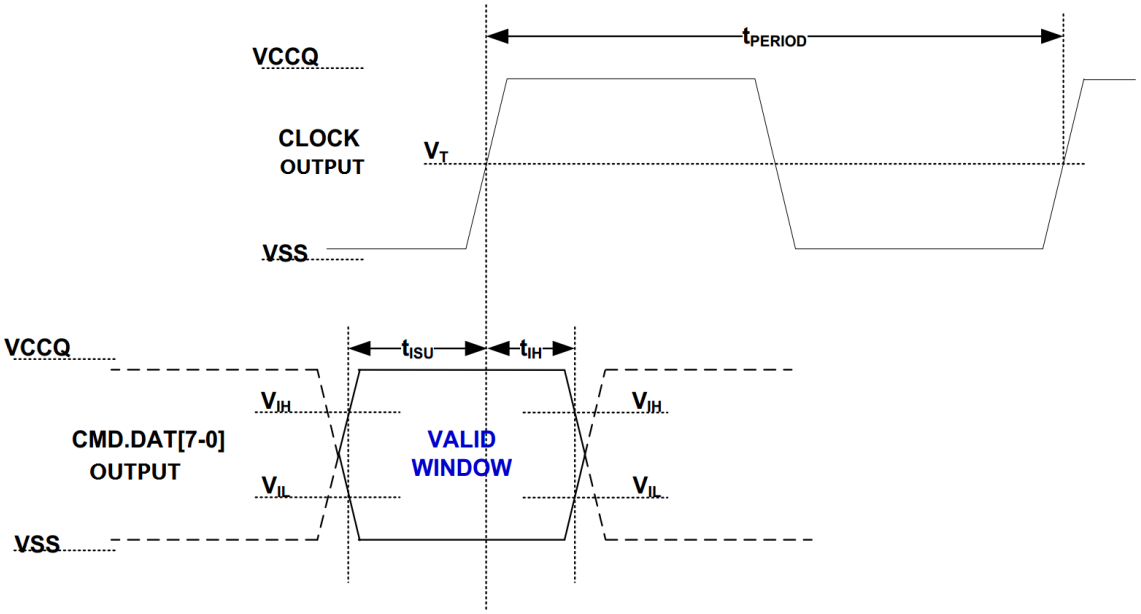
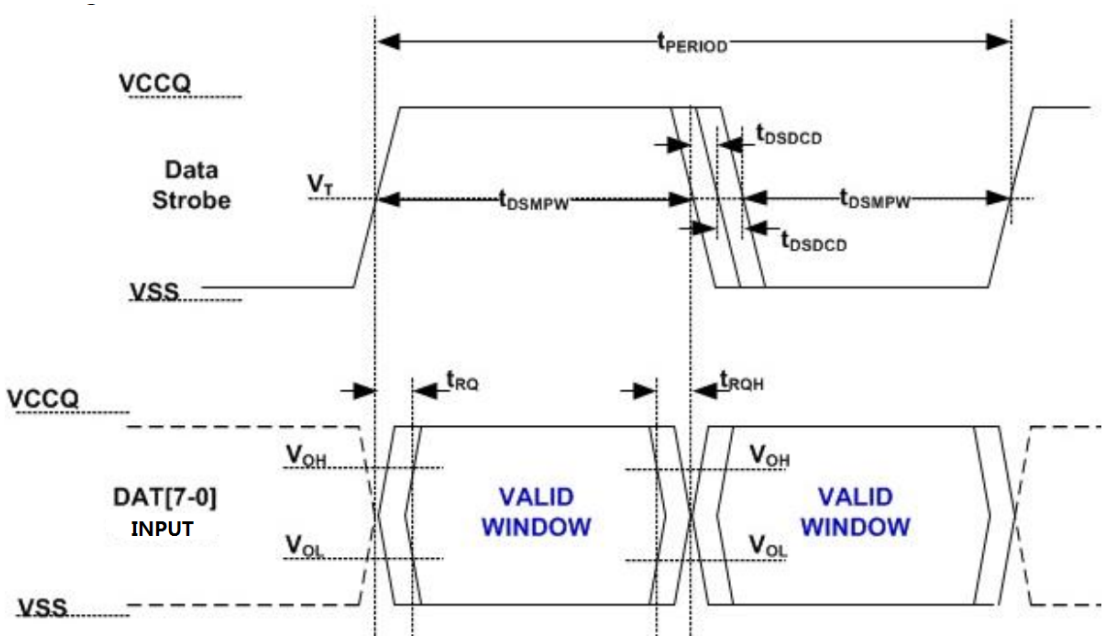


Table 5-13 HS200 Output Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
tISU	output set-up time	1.4	-	ns
tIH	output hold time	0.8	-	ns

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Figure 5-4 EMMC HS400 Data Input Timing



NOTE $V_T = 50\%$ of V_{CCQ}, indicates clock reference point for timing measurements.

Table 5-14 HS400 Data Input Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
SR	Slew rate	1.125	-	V/ns
tCKDCD	Duty cycle distortion	0	0.2	ns
tCKMPW	Minimum pulse width	2	-	ns
tRQ	Input skew	-	0.4	ns
tRQH	input hold skew	-	0.4	ns

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Figure 5-5 EMMC HS200 Data Input Timing

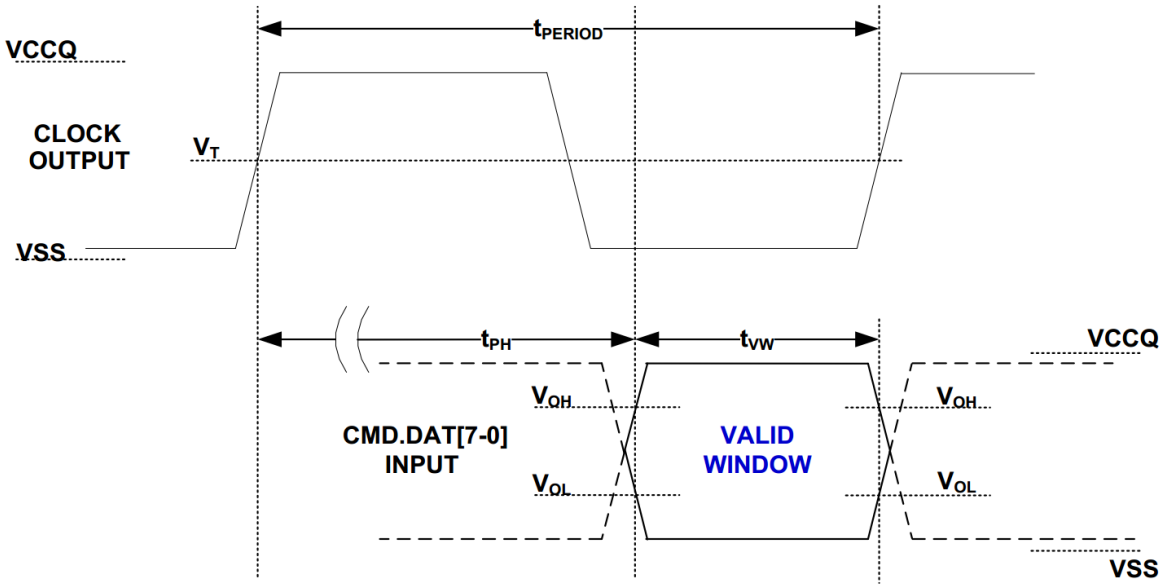
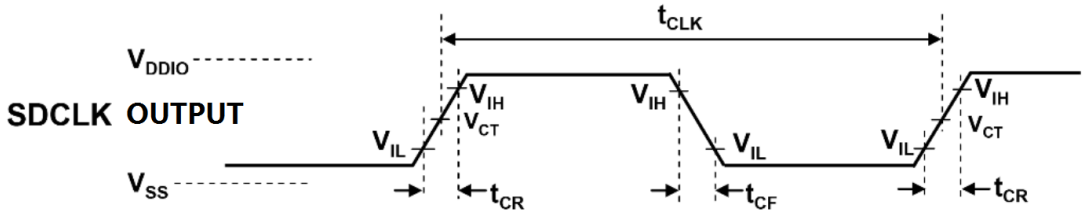


Table 5-15 HS200 Input Timing Specification

Symbol	Parameter	Min	Max	Unit
tPH	Device output momentary phase from CLK input to CMD or DAT line output. Does not include a long-term temperature drift.	0	2	UI
ΔTPH	Delay variation due to temperature change after tuning. Total allowable shift of output valid window (TVW) from last system Tuning procedure ΔTPH is 2600ps for ΔT from -25 °C to 125 °C during operation.	-350(ΔT=-20deg.C)	1550(ΔT=90deg.C)	ps
tVW	Valid Data Simple window	0.575	-	UI

Figure 5-6 SDIO(SDR104) Clock Signal Timing



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Table 5-16 SDIO(SDR104) Clock Timing Specification

Symbol	Parameter (SDR104 Mode)	Min	Max	Unit
tCLK	clock period Data Transfer Mode(PP)	4.8	-	ns
Duty	Clock Duty	30	70	%
tCR	clock rise time	-	0.96	ns
tCF	clock fall time	-	0.96	ns

Figure 5-7 SDIO(SDR104) Output Timing Diagram

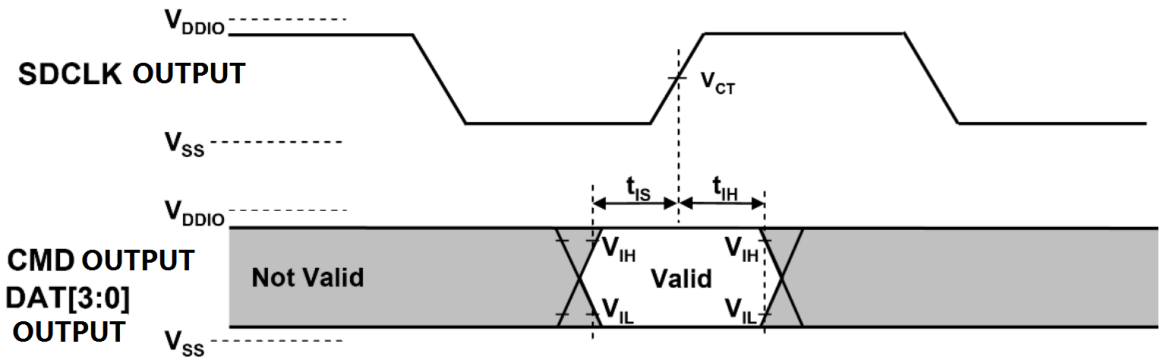


Table 5-17 SDIO(SDR104) Timing Specification

Inputs CMD, DAT (referenced to CLK)				
Symbol	**Parameter**	**Min**	**Max**	**Unit**
tIS	input set-up time	1.4	-	ns
tIH	input hold time	0.8	-	ns

Note

SD card interface uses SDIO protocol.

5.6.3 NAND Timing Specification

Nand timing specifications are shown as below.

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Figure 5-8 Async Waveform for Command/Address/Data Output Timing

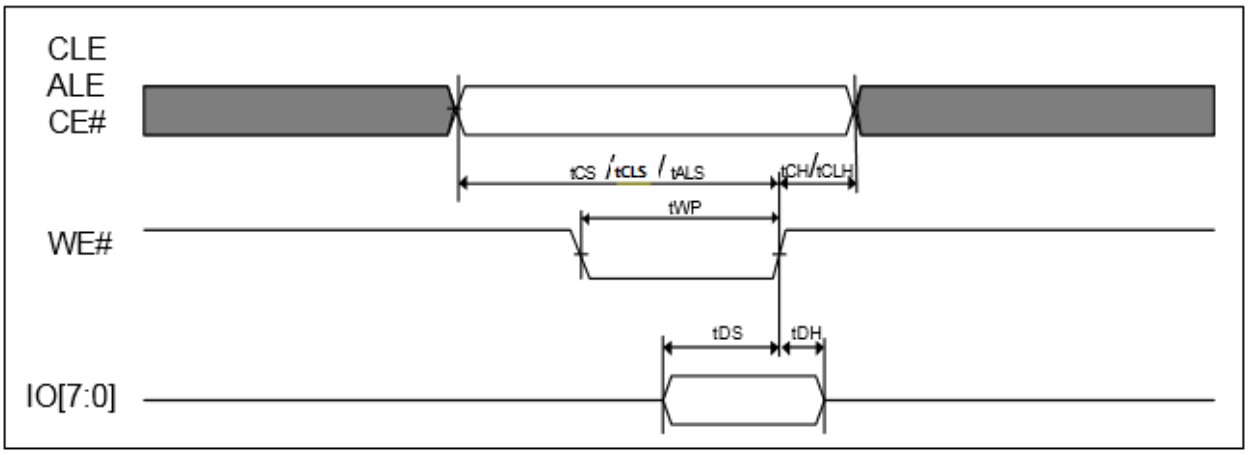
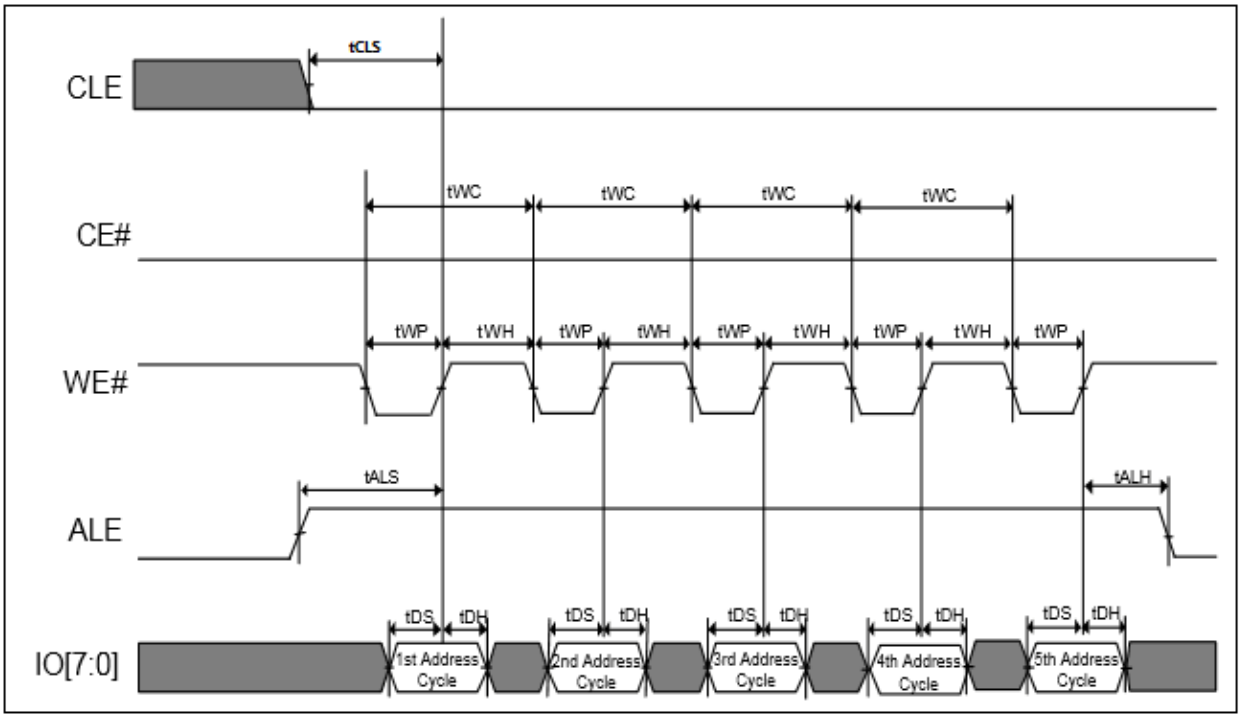


Figure 5-9 Async Waveform for Address Output Cycle



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Figure 5-10 Async Waveform for Sequential Data Read Cycle(After Read)-EOD Mode

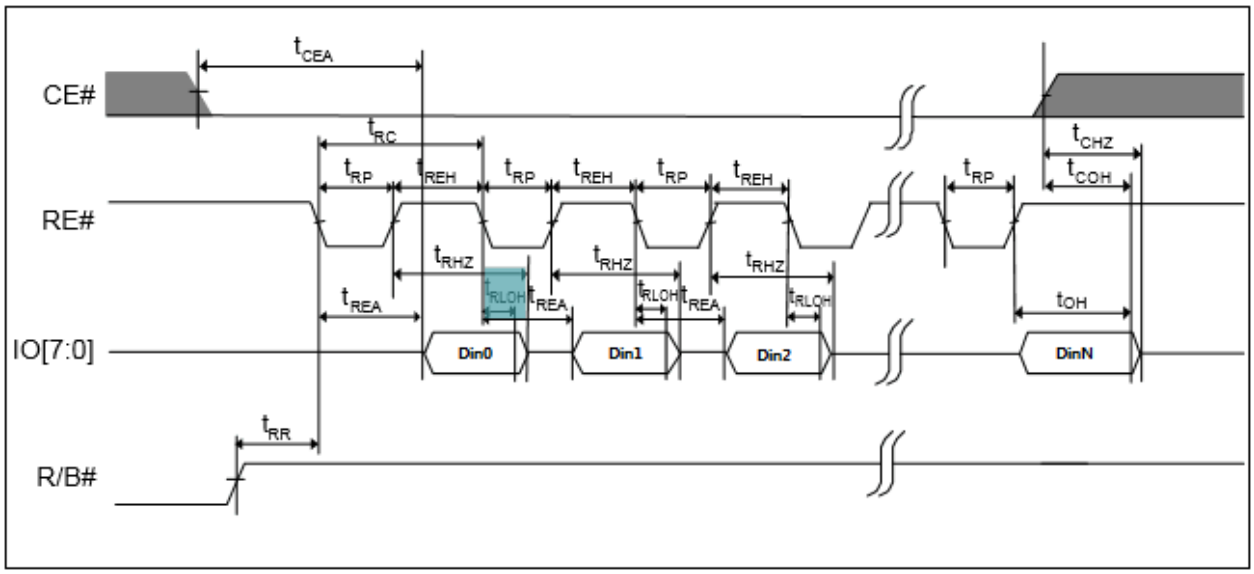


Table 5-18 Nand Timing Specifications

Symbol	Parameter (Asynchronous) (mode 5)	Min	Max	Unit
tCLS	CLE setup time	10	-	ns
tCLH	CLE hold time	5	-	ns
tALS	ALE setup	10	-	ns
tALH	ALE hold	5	-	ns
tDS	Data setup time	7	-	ns
tDH	Data hold time	5	-	ns
tWC	WE# cycle time	20	-	ns
tWP	WE# pulse width	10	-	ns
tWH	WE# high lold time	7	-	ns
tREA	RE# access time	-	16	ns
tOH	Data output hold time	15	-	ns
tRLOH	RE#-low to data hold time (EDO)	5	-	ns
tRP	RE# pulse width	10	-	ns
tREH	RE# high hold time	7	-	ns
tRC	RE# cycle time	20	-	ns

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5.6.4 SPICC Timing Specification

Figure 5-11 SPICC Timing Diagram

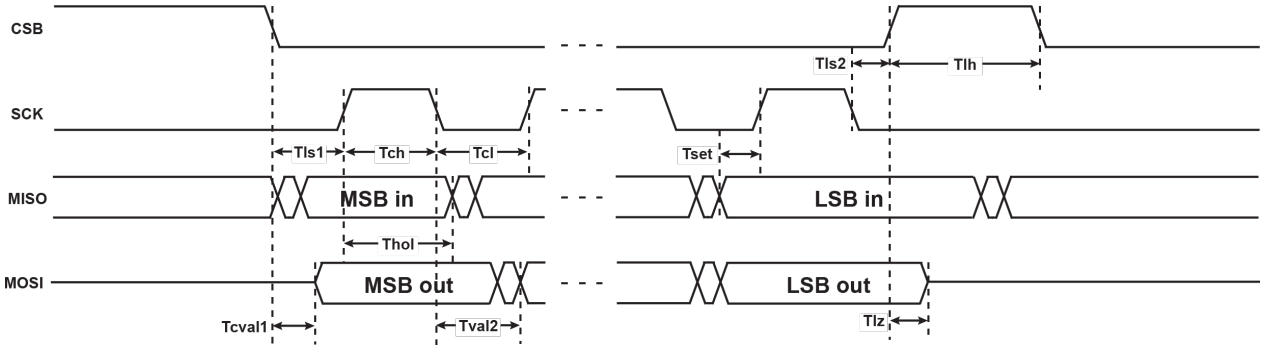
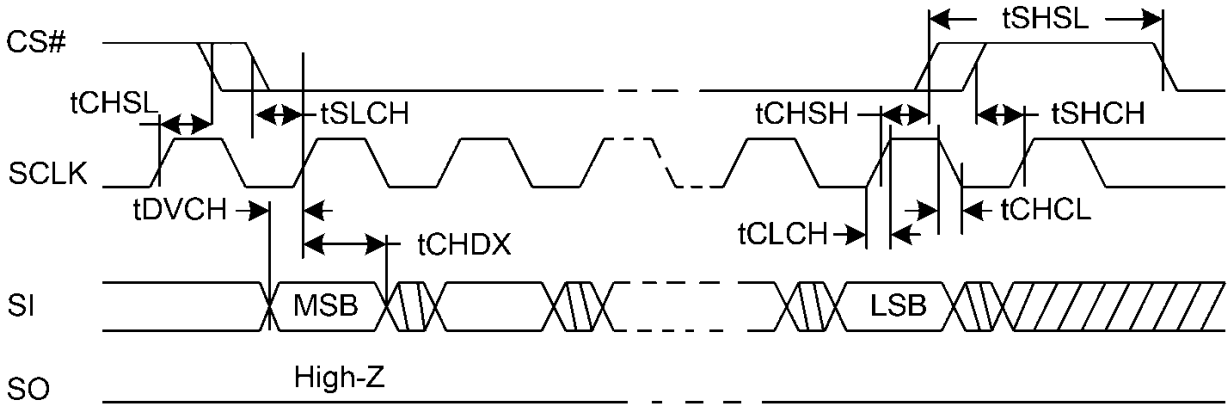


Table 5-19 SPICC Master Timing Specification

Symbol	Description	Min.	Max.	Unit
fCLK	Clock Frequency	1	80	MHz
TCH	Clock high time	5		ns
TCL	Clock low time	5		ns
TLS1	CS fall to First Rising CLK Edge	50		ns
TSET	Data input Setup Time	4		ns
THOL	Data input Hold Time	4		ns
TLH	Minimum idling time between transfers (minimum ss high time)	5		ns

5.6.5 SPIFC Timing Specification

Figure 5-12 SPIFC Serial Input Timing Diagram



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Figure 5-13 SPIFC Out Timing Diagram

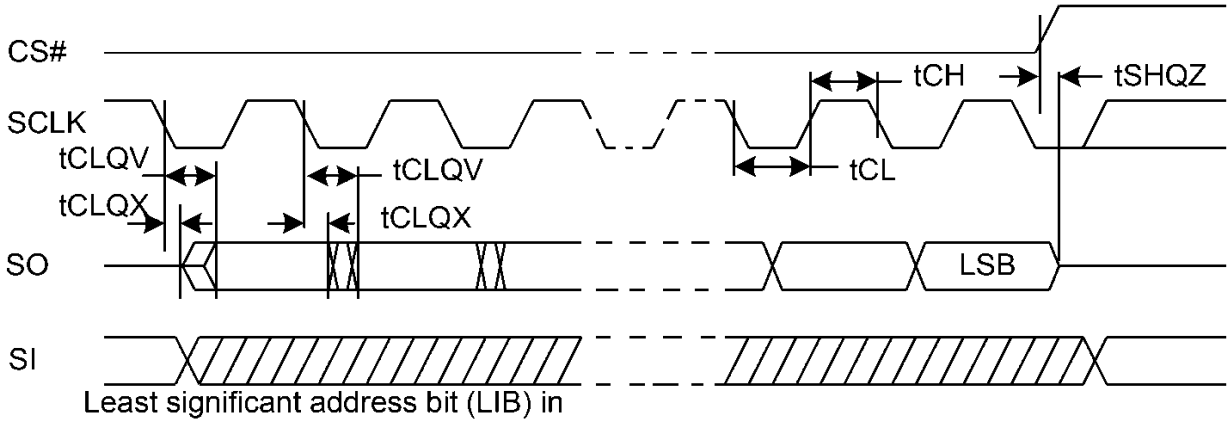


Table 5-20 SPIFC Master Timing Specification

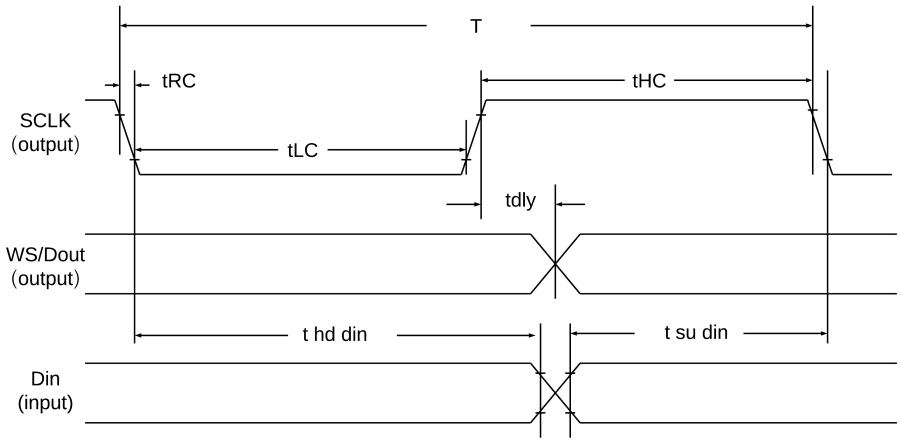
Symbol	Parameter (Clock 41.7MHz)	Min	Max	Unit
fRSCLK	Clock Frequency for READ instructions	-	50	Mhz
tCH	Clock High Time	8	-	ns
tCL	Clock Low Time	8	-	ns
tCLCH	Clock Rise Time (peak to peak)	0.1	-	V/ns
tCHCL	Clock Fall Time (peak to peak)	0.1	-	V/ns
tSLCH	CS# Active Setup Time (relative to SCLK)	4	-	ns
tCHSH	CS# Active Hold Time (relative to SCLK)	4	-	ns
tDVCH	Data In Setup Time	2	-	ns
tCHDX	Data In Hold Time	3	-	ns
tSHQZ	Output Disable Time (relative to CS#)	-	8	ns
tCLQV	Clock Low to Output Valid	-	6	ns
tCLQX	Output Hold Time	1	-	ns

5.6.6 Audio Timing Specification

There are two modes for the audio I2S/TDM interface: Master mode and Slave mode, as shown below.

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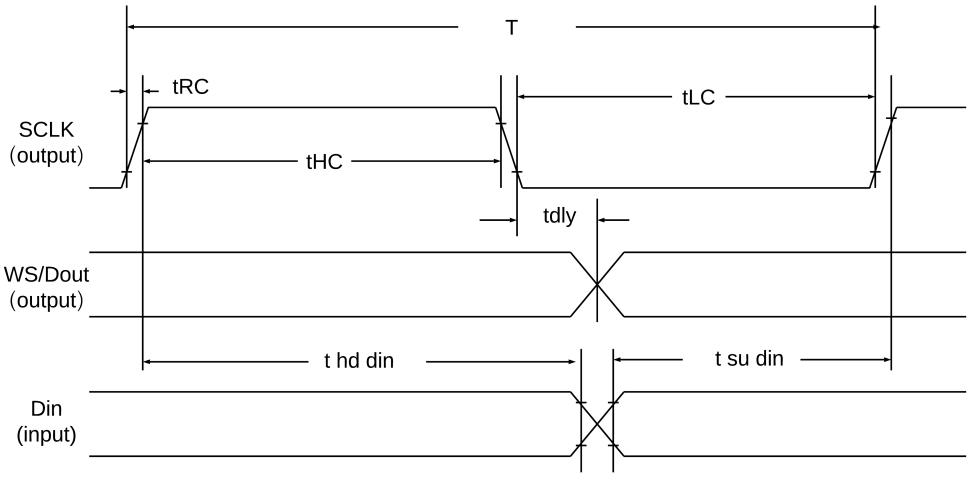
Figure 5-14 Output Data of SCLK Rising Edge TDM/I2S Timing, Master Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Figure 5-15 Output Data of SCLK Falling Edge TDM/I2S Timing, Master Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Table 5-21 Audio I2S/TDM Timing Specification, Transmitter, Master Mode

Symbol	Parameter	Min	Typ	Max	Unit
T	Clock period	10			ns
tHC	High level of SCLK	0.35			T
tLC	Low level of SCLK	0.35			T
tRC	Edge time of SCLK			0.15	T
tdly	Delay from SCLK to WS	-2	3	5	

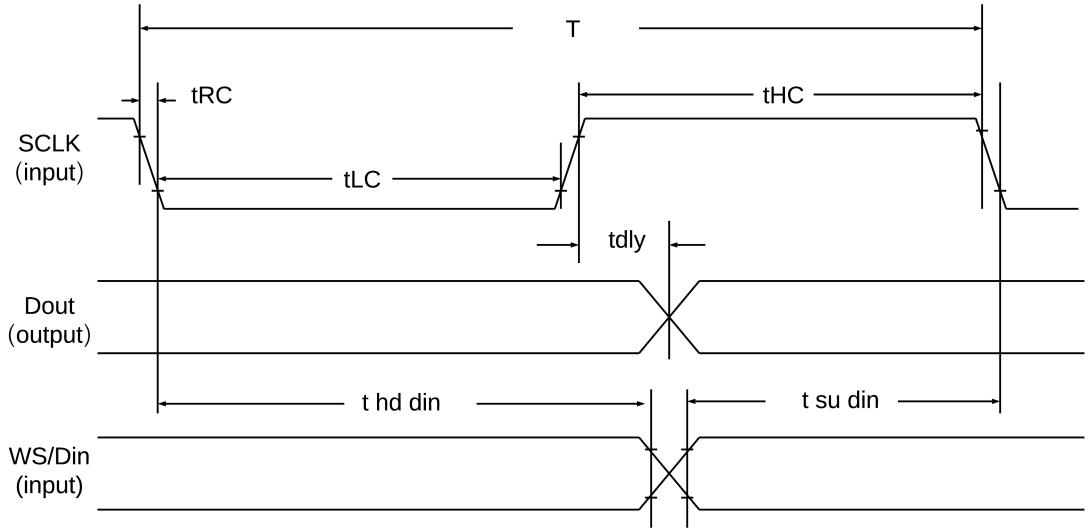
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Symbol	Parameter	Min	Typ	Max	Unit
tsuin	Setup time of Din	4			ns
thdin	Hold time of Din	4			ns

Note

Measure Point refers to VIH, ViL parameters of Normal GPIO Specifications.

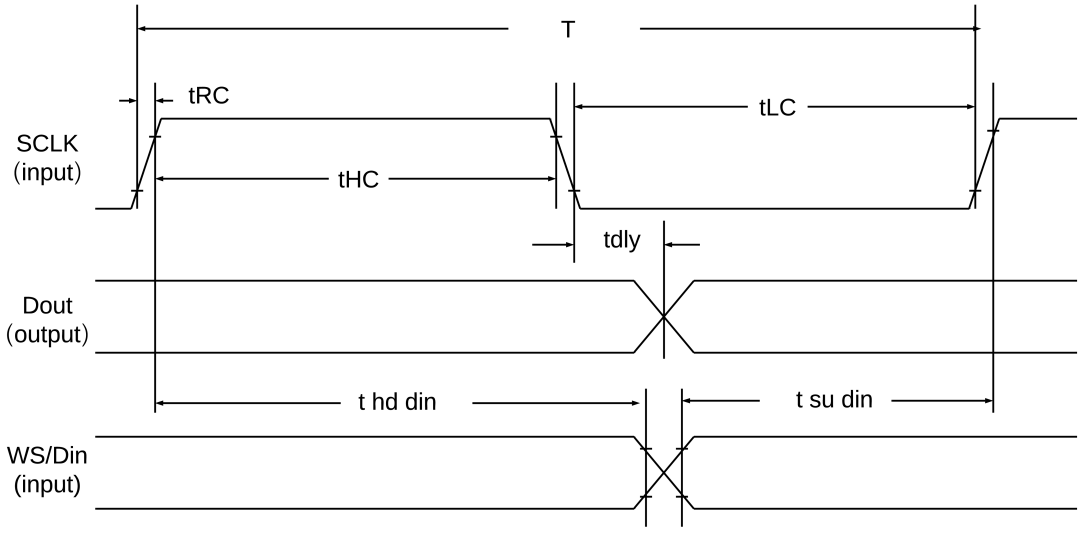
Figure 5-16 Output Data of SCLK Rising Edge TDM/I2S Timing, Slave Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Figure 5-17 Output Data of SCLK Falling Edge TDM/I2S Timing, Slave Mode



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Note

Input data was sampled on SCLK rising/falling edge via software setting.

Table 5-22 Audio I2S/TDM Timing Specification, Transmitter, Slave Mode

Symbol	Parameter	Min	Typ	Max	unit
T(out)	Clock period	40			ns
T(in)	Clock period	10			ns
tHC	High level of SCLK	0.35			T
tLC	Low level of SCLK	0.35			T
tRC	Edge time of SCLK			0.15	T
tsu din	Setup time of WS/Din	4			ns
thd din	Hold time of WS/Din	4			ns
tdly	Delay between SCLK and Dout	2	12	15	ns

Note

Measure Point refers to VIH, ViL parameter of Normal GPIO Specifications.

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5.6.7 PDM Timing Specification

Figure 5-18 PDM Timing Diagram

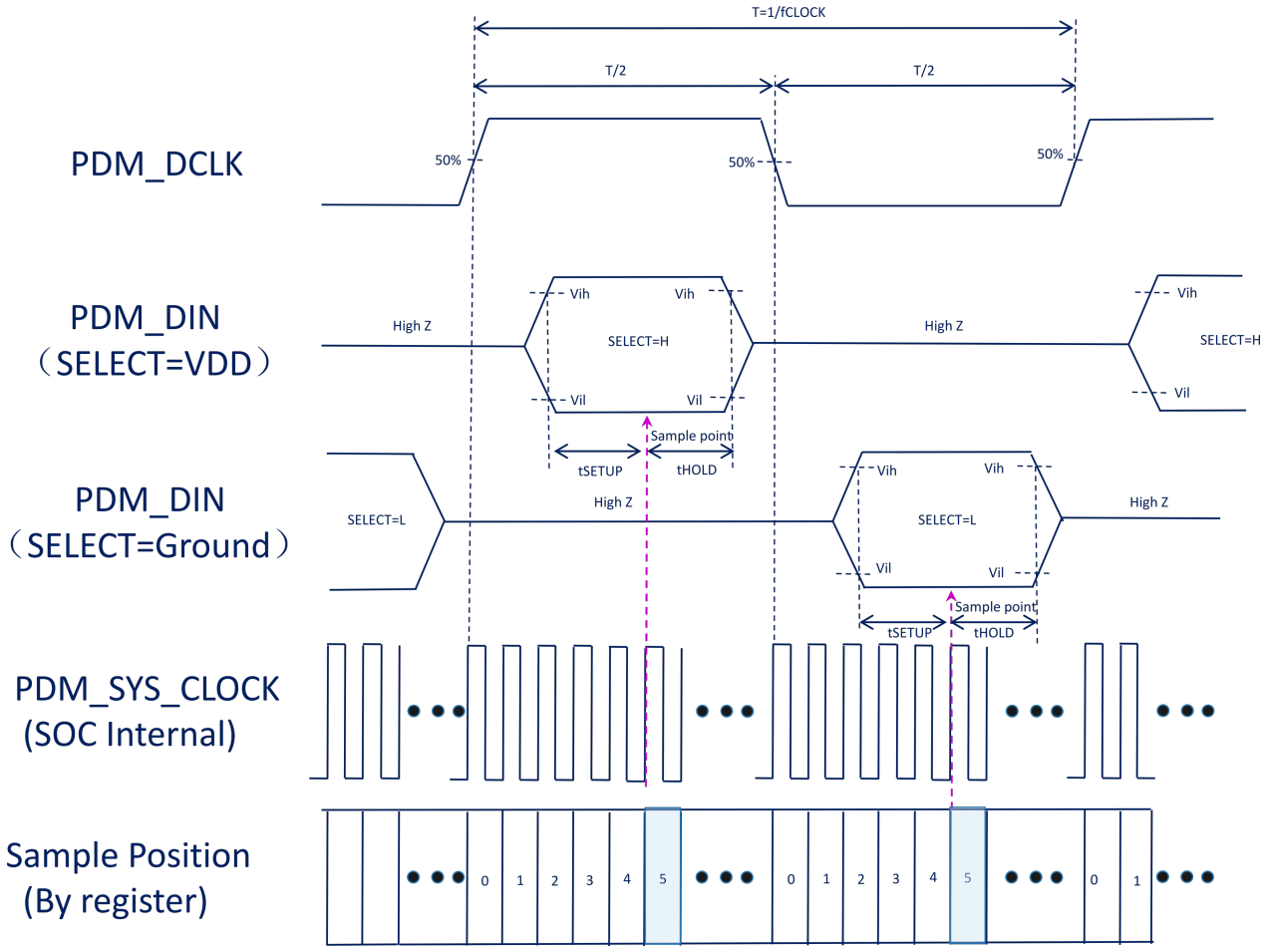


Table 5-23 PDM Timing Specification

Parameter	Symbol	Min.	Typ.	Max.	Unit
PDM clock period	tDCLK	200			ns
PDM clock duty cycle	tHIGH/tLOW	48%		52%	tDCLK
PDM Data setup time	tSETUP	20			ns
PDM Data hold time	tHOLD	20			ns
Sys clock period	tSYSCLK	5	7.5		ns

Note

1. Default PDM_SYS_CLOCK=133MHz.
2. For Sample position, please refer to PDM register PDM_CHAN_CTRL, PDM_CHAN_CTRL1.

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5.6.8 UART Timing Specification

Figure 5-19 UART Timing Diagram

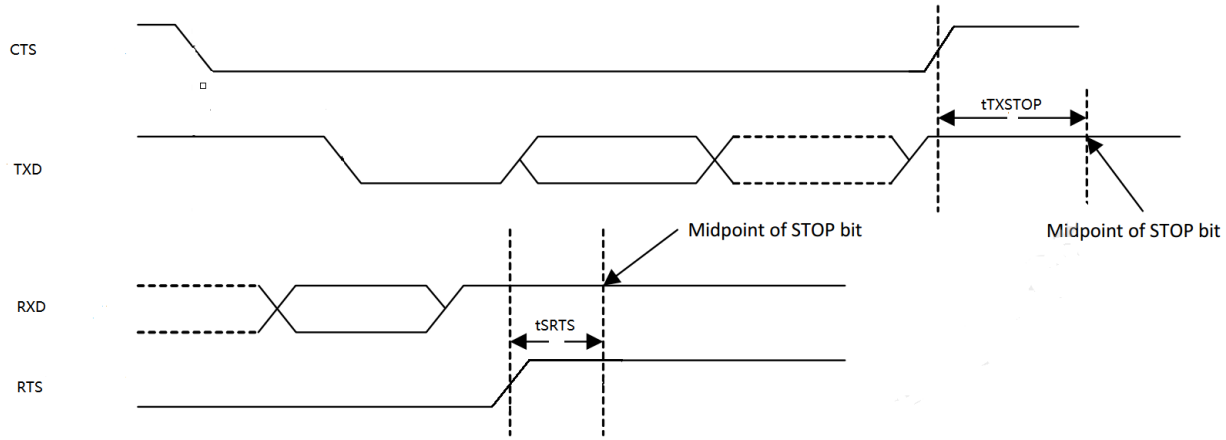


Table 5-24 UART Timing Specification

Parameter	Symbole	Min.	Max.	Unit
Delay time, CTS high before midpoint of stop bit	tTXSTOP	-	0.5	Bit Periods
Delay time, midpoint of stop bit to RTS hgih	tSRTS	-	0.5	Bit Periods

5.7 Recommended Oscillator Electrical Characteristics

The processor requires the 24MHz oscillator for generating the main clock source.

Table 5-25 24MHz Oscillator Specification

Symbol	Description	Min.	Typ.	Max.	Unit	Note
F_o	Nominal Frequency		24		MHz	
$\Delta f/f_o$	Frequency Tolerance	-30		30	ppm	At 25°C
		-50		50	ppm	At -20~85 °C
C_L	Load Capacitance	7.5	12	12.5	pF	
ESR	Equivalent Series Resistance			100	oHm	

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Note

- 10ppm Tolerance is preferred if 24MHz XTAL is also driving WIFI module.
- For user external clock source□please connect input clock output to SYS_OSCIN, let SYS_OSCOUT float.
- The threshold of Xin inverter is around 0.9V (Xin range: -0.3V to +2.1V). Therefore, follow the suggestions below for input clock.
  - Without DC blocking capacitor, use a higher Vpp output TCXO. The high voltage should be higher than 1.35V (VSWING >1.35V, 0V to >1.35V).
  - With DC blocking capacitor, re-bias the middle voltage at 0.9V, VSWING >2*0.45V.

5.8 Power On Config

3 Boot pins are used as power on config (POC) pins, to set the booting sequence.

POC setting is latched at the rising edge of reset signal.

3 POC pins are all pull high internal, CPU will try to boot from nand/eMMC first, if fails then try to boot from SD CARD, if still fails then try to boot from USB (PC).

External 4.7K ohm pull down resistors can be used to change the POC setting. The resistors should be placed on right location, avoid stubs on high speed signals.

The Power on configuration is listed below.

Table 5-26 Power On Configuration Pin Table

POC	Boot Pin	Name	Pull Low	Pull High
POC0	Boot [4]	SPI NAND First	SPI NAND boot first	Default sequence
POC1	Boot [5]	USB First	USB boot first	Default sequence
POC2	Boot [6]	SPI NOR First	SPI NOR first	Default sequence

Table 5-27 Booting Sequence Diagram

No.	POC_0 (SPI_ NAND)	POC_1 (USB_ BOOT)	POC_2 (SPI_ NOR, eMMC/ NAND)	1 st Boot device	2 nd Boot device	3 rd Boot device	4 th Boot device
1	0	0	0	USB (short delay)	SPI_NOR	NAND/eMMC	SD Card
2	0	0	1	USB (short delay)	NAND/ eMMC	SD Card	-
3	0	1	0	SPI_NOR	NAND/ eMMC	SD Card	USB
4	0	1	1	SPI_NAND	NAND/ eMMC	USB	-
5	1	0	0	USB (short delay)	SPI_NOR	NAND/eMMC	SD Card
6	1	0	1	USB (short delay)	NAND/ eMMC	SD Card	-

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No.	POC_0 (SPI_ NAND)	POC_1 (USB_ BOOT)	POC_2 (SPI_ NOR, eMMC/ NAND)	1 st Boot device	2 nd Boot device	3 rd Boot device	4 th Boot device
7	1	1	0	SPI_NOR	NAND/ eMMC	SD Card	USB
8	1	1	1	NAND/ eMMC	SD Card	USB	-

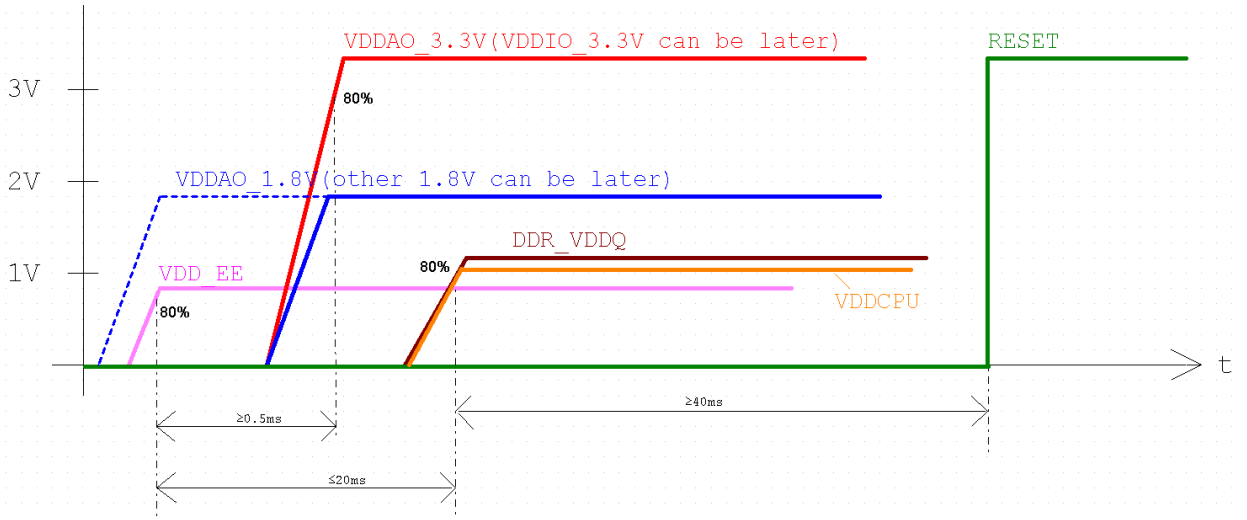
Note

If GPIOC is not working as SDCARD port, do not pull CARD_DET(GPIOC_10) low when system is booting up, to avoid romcode trying to boot from SDCARD.

5.9 Recommended Power On Sequence

The example of power on sequence is shown in the following figure.

Figure 5-20 Power On Sequence



Note

1. All test values refer to 80% of typical power voltage.
2. VDDAO_3.3V & VDDIO_3.3V should ramp up > 0.5ms later than VDD_EE.
3. All power sources should get stable within 20ms (except for DDR_VDDQ).
4. No sequence requirement between VDDCPU & DDR_VDDQ and other power source.
5. VDDAO_1.8V should ramps up no later than VDDAO_3.3V & VDDIO_3.3V, and VDDAO_ 3.3V & VDDIO_3.3V should never be 2.5V higher than VDDAO_1.8V.
6. In some designs, VDDCPU & VDD_EE are merged to VCC_CORE, the power on sequence should be same as VDD_EE.
7. RESET_n should keep low for at least 40ms after power up (except DDR_VDDQ).

Please refer to reference schematics.

5.10 Power Consumption

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Note

The specs listed here are tested typical value. Enough margin in circuit needs to be reserved.

Symbol	Maximum Current (mA)	Note
VDDCPU	2500	-
VDD_EE	2000	-
VDDQ	1600	Total power of controller and DDR3 memory

Symbol	Typical Current (mA)	Maximum Current (mA)	Note
AVDD08_P2P	TBD	200	
AVDD08_HDMIRX	80	110	
AVDD0V8_USB	25.8	39	Per USB channel at high speed full loading
VDD18_AO	2.3	-	Including VDD18_EFUSE Max 100 when programing EFUSE
AVDD18_DDR-P2P-DPLL	40	52	
AVDD18_ETH	33.2	50	-
AVDD18_AUDIO	12.3	-	-
AVDD18_V	56	110	
AVDD18_ADC_DPLL	5	7	-
AVDD18_HDMIRX	35	40	At 6Gbps mode
AVDD18_P2P_HRX-EARC	128	158	
AVDD18_USB	12.4	15	Per USB channel at high speed full loading
AVDD33_USB	5	29	Per USB channel at full speed full loading
AVDD33_HDMIRX	56	90	Per connected channel
VDDIO	-	-	[1]

Note

VDDIO=1.8V, DS=3, output 200MHz clock:

- IO pad itself consumes about 1.4mA.
- Driving a 55ohm trace with length of 50mm and width of 0.1mm will consumes about 2.8mA additional current (low impedance trace consumes more power)
- Base #2, add 5pF cap will consumes about 1.8mA additional current, total about 6mA
- When VDDIO=3.3V, GPIO consumes about 70% higher current, about 13mA
- Internal & external pull down resistor consumes additional current

5.11 Storage and Baking Conditions

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The processor is moisture-sensitive device of MSL level 3, defined by IPC/JEDEC J-STD-020. Please follow the storage and backing guidelines.

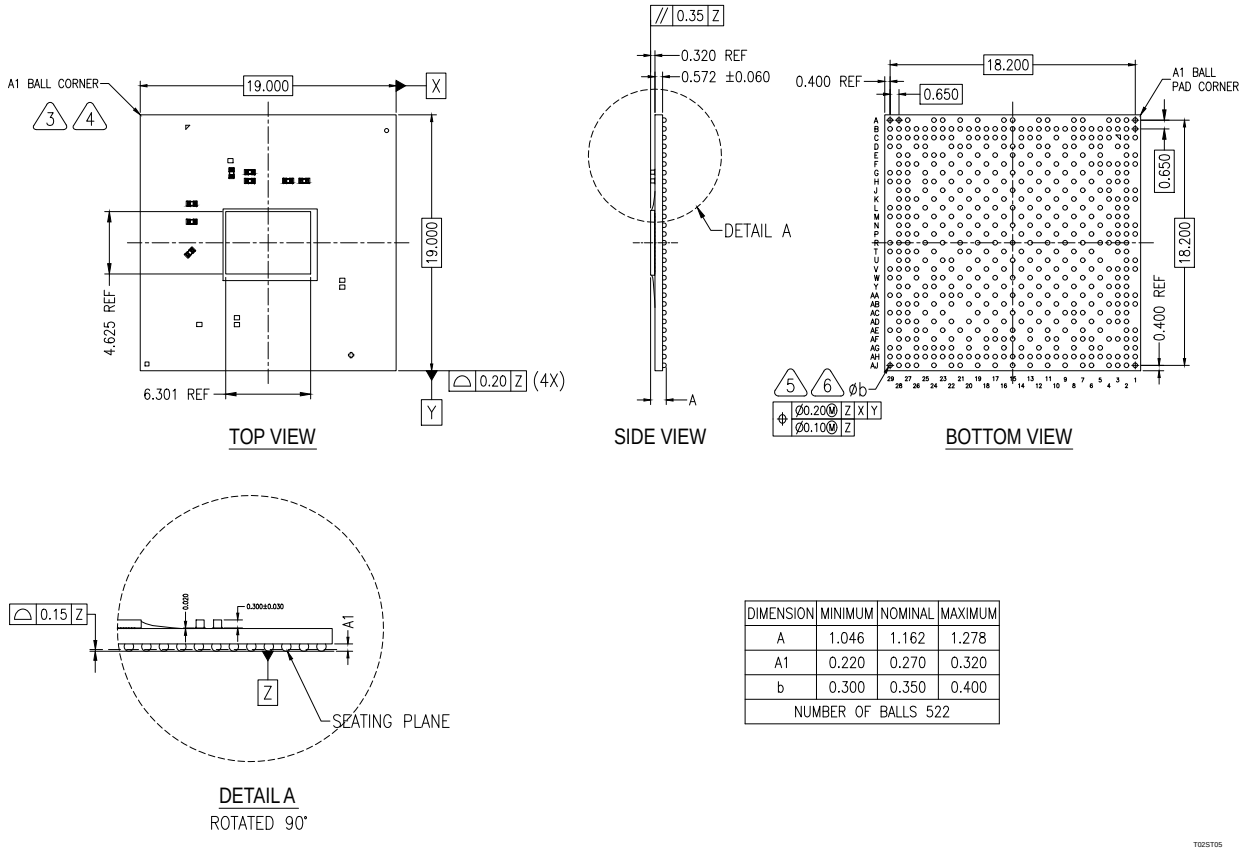
1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH).
2. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must comply with either of the following principles.
   a. Mounted with 168 hours of factory conditions ≤30°C/60% RH
   b. Stored per J-STD-033
3. If the humidity indicator card reads >10%, devices should be baked before mounting.
4. If baking is required, see IPC/JEDEC J-STB-033 for baking process.

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6 Mechanical Dimensions

The processor comes in a 29x29 ball matrix FCBGA or FCCSP RoHS package. The mechanical dimensions are given in millimeters as the following figure.

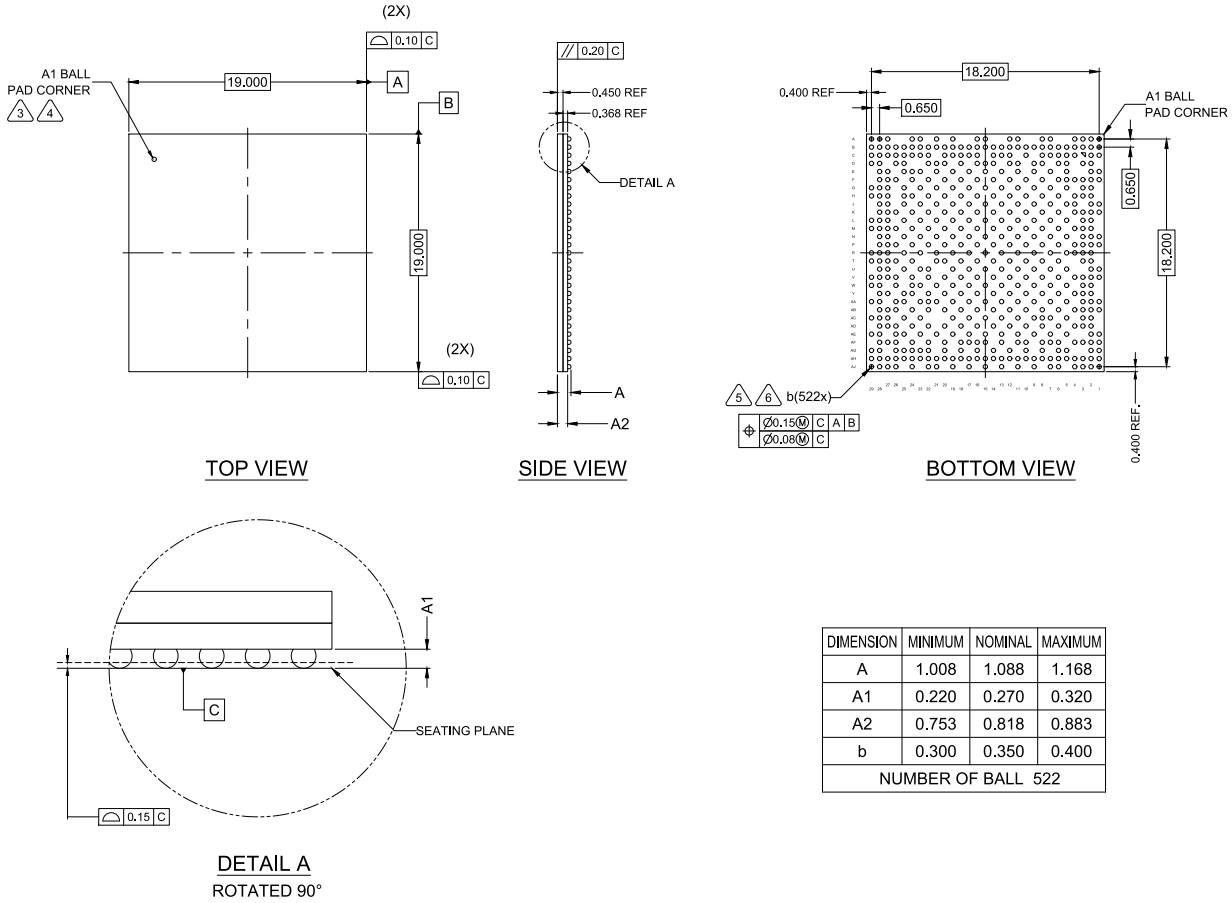
Figure 6-1 Dimensions (FCBGA)



DIMENSION	MINIMUM	NOMINAL	MAXIMUM
A	1.046	1.162	1.278
A1	0.220	0.270	0.320
b	0.300	0.350	0.400
NUMBER OF BALLS 522			

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Figure 6-2 Dimensions (FCCSP)



DIMENSION	MINIMUM	NOMINAL	MAXIMUM
A	1.008	1.088	1.168
A1	0.220	0.270	0.320
A2	0.753	0.818	0.883
b	0.300	0.350	0.400
NUMBER OF BALL 522			

T02ST10

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7 Appendix

7.1 DDR Length

The following table describes the DDR length of all nets on the SoC.

Table 7-1 DDR Length

Net Name	DDR3 Name	DDR4 Name	LPDDR4 Name	DDR Length(mm)
AC_0	CKE0	CKE0	CKEA0	7.32
AC_1	CKE1	CKE1	CKEA1	5.04
AC_2	CS_N0	CS_N0	CSA0	5.86
AC_3	NC	NC	CSA1	13.46
AC_4	BA2	ACT_N	CLKA_T	9.60
AC_5	RAS_N	WE_N	CLKA_C	10.25
AC_6	A7	BA0	NC	10.98
AC_7	A5	BG0	NC	11.32
AC_8	A10	A10	CAA1	9.13
AC_9	A14	A1	CAA0	13.95
AC_10	BA0	A8	CAA4	10.85
AC_11	CAS_N	A6	CAA5	10.20
AC_12	A9	A2	CAA2	13.73
AC_13	A13	A11	CAA3	7.39
AC_14	WE_N	A4	NC	7.65
AC_15	A11	A13	NC	12.14
AC_16	CLK0_T	CLK0_T	NC	6.93
AC_17	CLK0_C	CLK0_C	NC	7.63
AC_18	NC	NC	NC	8.91
AC_24	A6	A7	NC	9.03
AC_25	A4	A5	NC	9.08
AC_26	A1	A12	NC	6.86
AC_28	A8	BA1	NC	8.34
AC_29	BA1	RAS_N	NC	7.58
AC_30	A15	A0	NC	11.03
AC_31	A2	A9	NC	11.89
AC_33	A12	CAS_N	NC	8.48

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Net Name	DDR3 Name	DDR4 Name	LPDDR4 Name	DDR Length(mm)
AC_34	A3	BG1	NC	12.70
AC_35	A0	A3	NC	11.34
AC_36	ODT0	ODT0	NC	14.72
AC_37	ODT1	ODT1	NC	11.58
AC_38	CS_N1	CS_N1	NC	12.05
DDR_RSTN				6.15
DQ0				9.34
DQ1				9.85
DQ2				7.02
DQ3				5.57
DQ4				6.98
DQ5				7.83
DQ6				7.73
DQ7				7.66
DQ8				8.71
DQ9				9.17
DQ10				8.40
DQ11				8.87
DQ12				7.65
DQ13				5.86
DQ14				7.26
DQ15				8.76
DQ16				7.42
DQ17				6.18
DQ18				6.57
DQ19				4.77
DQ20				7.33
DQ21				7.61
DQ22				7.24
DQ23				4.04
DQ24				5.60
DQ25				5.02
DQ26				3.85
DQ27				6.98

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Net Name	DDR3 Name	DDR4 Name	LPDDR4 Name	DDR Length(mm)
DQ28				9.57
DQ29				9.66
DQ30				9.09
DQ31				9.91
DQM0				7.59
DQM1				8.79
DQM2				6.61
DQM3				7.16
DQS0_N				8.75
DQS0_P				7.78
DQS1_N				8.34
DQS1_P				7.78
DQS2_N				8.36
DQS2_P				7.71
DQS3_N				9.59
DQS3_P				8.97

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